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THE NATIONAL ENERGY COUNCIL

INDONESIA ENERGY **OUTLOOK** 2024



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FOREWORD BY MEMBER OF THE NATIONAL ENERGY COUNCIL

Let us praise to God the Almighty for His blessing that the National Energy Council has finished the book of Indonesia Energy Outlook 2024.

Indonesia Energy Outlook potrays the energy in Indonesia including the current energy condition and energy projection in the future.

The book of Indonesia Energy Outlook 2024 is jointly written by Members of the National Energy Council from Stakeholders and Editorial Board at the Secretariate General of the National Energy Council with the assistance from the experts. The formulation of Indonesia Energy Outlook also involves units in the Ministry of Energy and Mineral Resources namely Center for Data and Technology Information, Directorate General of Electricity, Directorate General of Oil and Gas, Directorate General of New Renewable Energy and Energy Conservation, as well as Directorate General of Mineral and Coal especially on the collection of current data and planning data.

In Energy Outlook 2024, the energy supply and demand projection are carried out for the next 10 years in 2 (two) scenarios namely Reference (REF) which depicts the deployment of the existing policy and GREEN scenario which uses assumptions toward 2045 Advanced Indonesia and Net Zero Emission (NZE) by 2060.

In addition to this 2024 publication, Indonesia Energy Outlook also presents the result of projection based on 7 (seven) regions namely Sumatra, Java-Bali, Kalimantan, Sulawesi, Nusa Tenggara, Maluku and Papua.

Jakarta, December 2024

Members of National Energy Council from Industrial Stakeholder

Abadi Purnomo

FOREWORD BY SECRETARY GENERAL OF THE NATIONAL ENERGY COUNCIL

The book of Indonesia Energy Outlook 2024 is an annual publication by the Secretariate General of the National Energy Council since 2014.

The energy supply and demand projection are conducted based on LEAP (Low Emissions Analysis Platform) modelling using 2023 as the baseline year taken from Center for Data and Technology Information Ministry of Energy and Mineral Resources and BPS (Indonesia Statistic Agency) as well as data from related units.

In the effort to improve the quality of energy supply and demand projection data, Secretariate General of the National Energy Council would enhance the relationship with related parties to publish the book of Indonesia Energy Outlook as a reliable and trusted reference.

Jakarta, December 2024

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- Members of the National Energy Council from stakeholders;
- Directorate General of Oil and Gas, Directorate General of Electricity, Directorate General of Mineral and Coal, Directorate General of New Renewable Energy and Energy Conservation, Center for Data and Technology Information MEMR and Geological Agency;
- Danish Energy Agency;
- Energy experts for their contribution in formulating Indonesia Energy Outlook 2024.

DISCLOSURE

Indonesia Energy Outlook 2024 is an analysis on the long-term national energy demand and supply projection (2024-2034) with particular assumptions which are developed for the purpose of formulating future energy projection scenario. The assumption and projection are based on energy technology development including fossil and renewable energy based on current data and condition. The data in Indonesia Energy Outlook is taken from official publication as well as temporary data or data that are continuously updated by the sources.

EXECUTIVE SUMMARY

The 2024 edition of Indonesia Energy Outlook develops energy modeling for 2024 - 2034 into 2 (two) scenarios, namely: the REF scenario which uses current policy assumptions, and the GREEN scenario which uses new policy assumptions towards an 2045 Advanced Indonesia and NZE in 2060. Projections are made in 7 (seven) regions, namely: Sumatra, Java-Bali, Kalimantan, Sulawesi, Nusa Tenggara, Maluku, and Papua, taking into account the geographical conditions of several adjacent provinces.

The energy supply and demand analysis are conducted based on calculation of LEAP (Low Emissions Analysis Platform) model as a simulation model of energy planning which can conduct an integrated energy supply and demand analysis. In LEAP model, energy demand projection is calculated based on the multiplication of energy consumption activity and energy consumption intensity. Energy activity is described by economic growth, population and others. Meanwhile, energy intensity is the level of energy consumption per unit of activity in particular period.

As input in modeling, the main data sources used in formulating the IEO 2024 are the Handbook of Energy and Economic Statistics of Indonesia (HEESI) 2023 published by Pusdatin MEMR (Center for Data and Technology Information), RUPTL (General Plan for the Supply of PLN Electricity), Statistik Indonesia 2023-BPS, and BPS Industrial Statistics data. Regional energy consumption is determined by summing data from each province following the division by region. Energy consumption data per province is obtained from units in MEMR and energy Business Entities. Specifically, oil and LPG consumption data are obtained from oil and LPG sales data per province from the Directorate General of Oil and Gas, for electricity consumption, the main data is based on PLN sales data, with additional identification of non-PLN consumption data and electricity from the smelter industry from the Directorate General of Electricity. For coal consumption data per province, it is obtained from the Directorate General of Mineral and Coal. Specifically, gas consumption data is based on sales data from HEESI, while consumption per industrial subsector uses BPS Industrial Statistics data. Household consumption data per province uses data from the Indonesia Residential End Use Survey which is the result of a collaborative study between CLASP and the Directorate General of NREEC.

The energy assumptions used in the REF scenario refer to current conditions and future projections based on historical data from the last few years, including the implementation of biofuels, electric vehicles, etc., while for the construction of power plants it refers to the RUPTL. In the GREEN scenario, the assumptions for

economic and population growth are the same as the REF scenario, but regarding energy utilization, assumptions are used that lead to decarbonization efforts towards NZE 2060, both in the household, transportation, industry, and power generation sectors. As a basic assumption, economic growth in both scenarios is 5.7%, while population growth is around 0.8% for the period 2023 - 2034.

Over the next 10 years, the REF scenario projects an average increase in final energy demand of 6.0% per year, while the GREEN scenario projects an increase of only around 4.7% per year due to the implementation of more intensive energy saving technologies and energy conservation in all sectors. A significant difference in growth between scenarios occurs in the transportation sector. In the REF scenario, the increase in final energy demand in the transportation sector is 5.7% in average, but in the GREEN scenario, there is only an increase of 2.6% in average per year, influenced by increased efficiency through the use of electric vehicles.

Based on the type of energy, oil will still dominate final energy consumption for both scenarios, 43% in the REF scenario and 31% in the GREEN scenario. However, the growth of oil consumption in the GREEN scenario shows below the average growth of total final energy consumption due to efforts to substitute oil for electric vehicles and the use of biofuels. The growth of NRE consumption increases the highest compared to other types of energy, reaching 8.2% per year (REF) and 9.4% per year (GREEN) respectively, mainly driven by an increase in the FAME mixture in diesel oil by 40% and ethanol in gasoline by 5% after 2030.

Based on the region, Java-Bali still dominates final energy consumption but with the smallest growth among other regions. In the REF scenario, the national average consumption growth is 6.0% per year, while the average consumption growth in Java-Bali is only 4.7% per year (REF) and 3.4% per year (GREEN). On the other hand, the Maluku region has a small consumption volume, but its energy consumption growth is projected to be the largest, reaching 9.7% per year (REF) and 14.9% per year (GREEN) driven by the growing mineral processing and refining industry.

Total electricity consumption is projected to increase from 315 TWh in 2023 to 547 TWh (REF) and 713 TWh (GREEN) in 2034 or experiencing an average growth of around 5.3% per year (REF) and 7.7% per year (GREEN). In line with the increase in electricity consumption, total electricity production is projected to increase to around 584 TWh (REF), and 712 TWh (GREEN) whose contribution still comes from fossil energy, especially coal. In the REF scenario, the share of electricity production from coal-fired power plants is projected to increase from around 56% in 2023 to 67% in 2034. Meanwhile, for the GREEN scenario, the share is

projected to decrease to 44%. On the other hand, electricity production from renewable energy power plants in the next ten years, especially in the GREEN scenario, has experienced quite high growth, namely 10% per year. Meanwhile, for the REF scenario, because it is still dominated by coal-fired power plants, production growth from renewable power plants only increases by an average of 3.1% per year. The largest increase in renewable electricity production in the GREEN scenario comes from solar power with an average growth of 49.3% per year. Thus, the production increases from 0.7 TWh in 2023 to 59 TWh in 2034, supported by the continued decline in the trend of solar power plant installation costs.

To meet electricity demand, power generation capacity is projected to increase from 91 GW in 2021 to 140 GW (REF) and 213 GW (GREEN) in 2034. All types of power plants experienced positive growth, except for Diesel Power Plant which continued to decline. NRE power plant capacity experienced the highest growth compared to other energy sources. Thus, NRE power plant capacity will increase from 14 GW in 2023 to 27 GW (REF) and 82 GW (GREEN) in 2034.

The total primary energy supply in 2034 is projected to increase by around 5.3% and 4.7% per year or to 456 million TOE (REF) and 429 million TOE (GREEN), respectively. The supply of NRE in 2034 in the REF scenario reaches 71 million TOE, or around 15.5% of the total primary energy supply, while in the GREEN scenario, the supply of NRE is around 112 million TOE, or 26.1% of the total primary energy supply.

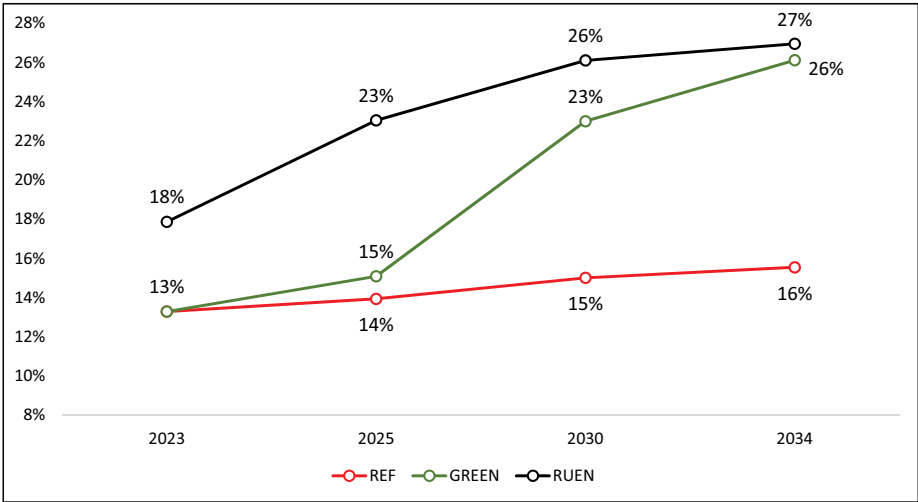
In terms of emissions produced, total emissions from the energy sector in 2034 are projected to increase to 1,359 million tons of CO₂eq (REF) and 1,105 million tons of CO₂eq (GREEN). With this target, the national commitment to emission reductions in the ENDC will be met. Throughout the projection period, power plants are still the largest contributor to emissions because the use of fossil fuels, especially coal, is still dominant at around 38% (REF) and 42% (GREEN) in 2034. Meanwhile, the industrial sector contributes 33% (REF) and 37% (GREEN). The high CO₂ emissions from industry are because there is still a lot of equipment in the production, cooling, and heating processes that still use fossil fuels and cannot be substituted by electricity.

Indonesia is a country with low electricity consumption per capita. Currently, Indonesia's average electricity consumption is 1,337 kWh per capita or far below the average of ASEAN countries of 3,896 kWh per capita. In 2025, electricity demand per capita in both scenarios (REF & GREEN) is 1,369 kWh/capita and 1,418 kWh/capita, respectively. It is still below the National Energy Policy target of 2,500 kWh/capita in 2025. Meanwhile, in 2034, electricity demand per capita will reach

around 2,505 kWh/capita in GREEN scenario and only 1,929 kWh/capita in REF scenario.

In 2023, Indonesia's final energy demand per capita is 0.64 TOE/capita, and in 2034 it is projected to increase to 1.11 TOE/capita (REF) and 1.00 TOE/capita (GREEN). This figure is lower compared to the global final energy per capita in 2022, which has reached 1.89 TOE/capita (EIA, 2024).

As an energy indicator that is often observed, the projection of NRE in the Total Primary Energy Supply in RUEN in 2025 is targeted at 25%. However, the achievement is only 13.9% in the REF scenario and 15.1% in the GREEN scenario as shown in the Figure of NRE mix projection 2023-2034 below.



NRE Mix Projection Year 2023 - 2034

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CHAPTER I

INTRODUCTION



CHAPTER I

INTRODUCTION

1.1 BACKGROUND

The book of IEO which is published annually is a result of an analysis that gives an overview of the national energy condition especially on energy supply and demand projection for the next 10 years (2024-2034).

The 2024-2034 projection uses two scenarios namely REF scenario which uses current policy assumptions and GREEN scenario which uses new policy assumptions toward 2045 Advanced Indonesia and Net Zero Emission (NZE) 2060.

The main data source in IEO 2024 is Handbook of Energy and Economic Statistics of Indonesia (HEESI) 2023 publication by Center for Data and Technology Information MEMR, RUPTL PLN, Statistics Indonesia 2023-BPS and Industrial Statistics of BPS.

1.2 METHODOLOGY

1.2.1 Modeling Analysis Framework

The modeling analysis is divided into three stages namely final energy demand, energy transformation, and primary energy supply. Final energy demand analysis is done by using the assumptions of GDP growth, population growth, policy, strategic planning and roadmap on current energy development. The primary

energy supply analysis also takes into account the use of various energy sources and energy resources potential including the current policies and current energy technology development. Meanwhile, energy transformation analysis is done by considering Electricity Supply General Plan (RUPTL). The modeling analysis framework is shown in Figure 1.1.

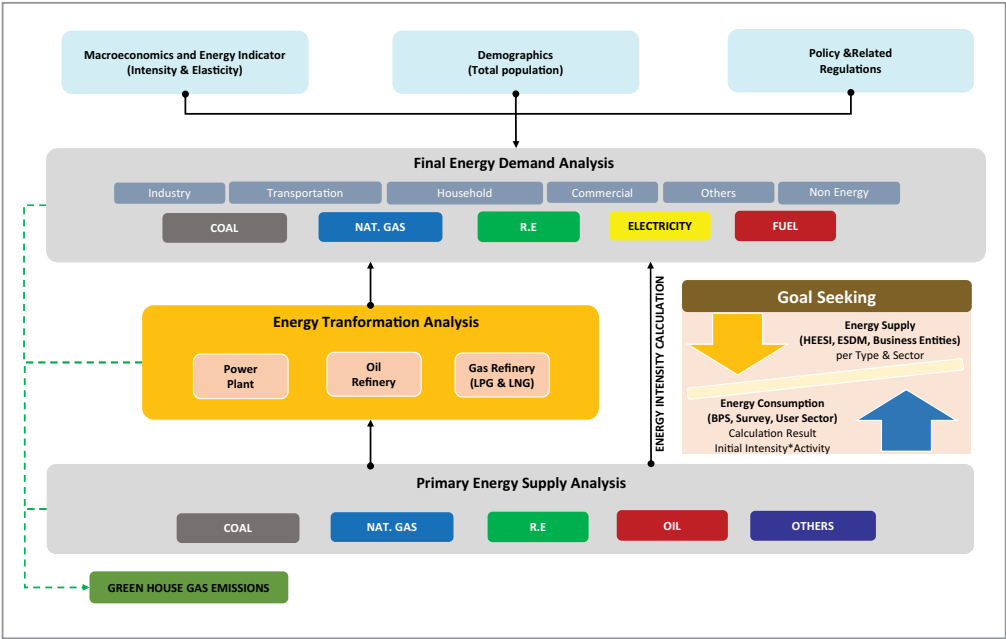


Figure 1.1 Modeling Analysis Framework

The energy supply and demand analysis are conducted based on the LEAP model; as a simulation model in energy planning which an integrated energy supply and demand analysis. In LEAP model, energy demand is projected based on the multiplication of energy consumption activity and energy consumption intensity. Energy activity is described by economic growth, population and others. Meanwhile, energy intensity is the level of energy consumption per unit of activity in particular period.

Based on the modeling analysis framework in Figure 1.1, the parameters in final energy demand projection are social economic data (population and economic growth), historical data on energy consumption to identify energy intensity and energy consumption pattern as the impact of people’s lifestyle that is influenced by the projection of GDP increase or a more efficient technology.

In this Indonesian Energy Outlook 2024, the energy supply demand projection by region will also be carried out to obtain the overview of energy demand and supply in each region. The projection is done in seven regions namely Sumatra, Java-Bali, Kalimantan, Sulawesi, Nusa Tenggara (Nusra), Maluku and Papua. The consideration in determining the seven regions is the geographical condition of several provinces located close to each other.

In the modeling, energy consumption uses the data on each province. It is then calculated based on by region. The data of energy consumption per province is obtained from units in MEMR and energy Business Entities, Statistical Offices (BPS), regional offices, etc, Data on oil and LPG consumption/sales per province is specifically derived from Directorate General of Oil and Gas. Meanwhile, the main data of electricity sales is derived from PLN sales data with additional identification of non-PLN consumption data and electricity in smelter industry from Directorate General of Electricity. Data on coal consumption per province is obtained from Directorate General of Mineral and Coal. However, the sales of electricity per industrial subsector refers to data from Industrial Statistics BPS. Specifically for data on gas consumption is based on data on sales at HEESI, while consumption per industrial subsector uses Industrial Statistics BPS data. The data on household consumption per province uses data from Indonesia Residential End Use Survey as the result of the joint study by CLASP and Directorate General of New Renewable Energy and Energy Conservation.

1.2.2 Energy Projection Scenario

The economic growth assumption used in both scenarios is around 5.7% and the population growth assumption in the 2023-2034 period is around 0.8%.

1.2.2.1 Reference (REF) Scenario

National economic growth used in the projections comes from BPS, LPEM UI and Bappenas 2024. These assumptions are also used to project energy demand in the current National Energy Policy Revision.

For the projection of economic growth by province, the GRDP Growth Projection data for 34 Provinces 2021-2035 from Bappenas is used. The regional economic growth will be the summation of the provinces in the region.

Meanwhile, the population assumptions is based on the BPS publication (2020 Population Census) covering the 34 provinces. Similar to economic growth, the regional population growth will be the summation of the provinces in the region.

The energy assumptions used in the REF scenario refers to the current conditions and future projections which was based on historical data from the last few years, including the implementation of biofuels, electric vehicles, etc., The construction of power plants refers to the RUPTL.

1.2.2.2 GREEN Scenario

The assumptions of economic and population growth in the GREEN scenario are the same as in the REF scenario. The energy utilization uses the assumptions that lead to the Advanced Indonesia in 2045 and NZE in 2060. This covers amongst other the following:

- a. Household sector: the use of city gas networks and electric stoves is projected to grow higher than in the REF scenario, as well as the conservation of equipment use in the household sector.
- b. Transportation sector: the use of EV and biofuel is projected to be higher than in the REF scenario (B40 in 2030 and E5 in 2033) as well as conservation in the transportation sector.
- c. Industrial sector: gradual substitution of the use of diesel to electricity and industrial conservation that was not carried out in the REF scenario.
- d. Power plant: Increasing NRE power plant capacity as compared to REF, especially for Solar Power Plant, Wind Power Plant. In addition, Coal Fired Power Plant is higher than in the REF scenario (15% in 2030) and the introduction of Nuclear Power Plant from 2032 onwards.

1.3 Report Outline

The outline of the Book Indonesia Energy Outlook 2024 is as follows:

CHAPTER I INTRODUCTION

This chapter includes brief explanation on the purpose and objective of the Book Indonesia Energy Outlook 2024, methodology, scenario, and assumption in modeling.

CHAPTER II ENERGY OUTLOOK 2024-2034

This chapter provides Indonesia energy condition projection for the next 10 years and the result of the analysis which include: final energy consumption, energy in electricity, primary energy supply, and energy indicator. This chapter also discusses the relation between current achievement and primary energy mix target based on Presidential Regulation on National Energy General Plan.

CHAPTER III PROGRESS OF ENERGY TRANSITION

This chapter presents the progress of the implementation toward a number of national policies that are mainly related to related to Indonesia Energy Transition as the focus.

CHAPTER IV CONCLUSION AND RECOMMENDATION

This chapter presents summary of the analysis on energy supply and demand projection in the next 10 years.

CHAPTER II

ENERGY OUTLOOK 2024-2034



CHAPTER II

ENERGY OUTLOOK 2024-2034

2.1 FINAL ENERGY CONSUMPTION

2.1.1 Final Energy Consumption by Sector

In general, energy demand is greatly influenced by several variables, including economic growth, population growth, technology penetration, energy prices, infrastructure availability and government policies. During the period 2010-2023, final energy consumption has increased by 5% per year or increased from 94 million TOE (2010) to 178 million TOE (2023). In 2023, the share of final energy consumption in the industrial sector was around 46%, followed by the transportation sector at around 35%. Meanwhile, energy consumption in the household sector was around 13%, commercial was around 4% and other sectors (agriculture, mining and construction) was around 1%. However, if we look at the growth in energy use, the household sector experienced the highest increase (an average of 5.6% per year), followed by the transportation sector (5.3% per year). Meanwhile, the commercial sector experienced the lowest increase in energy consumption (5.0% per year). The increase in energy use in the transportation sector is driven by rapid urbanization, resulting in an increase in the need for transportation to support mobility and increased economic activity, demand for transportation for the distribution of goods and worker mobility.

In the next ten years, without new intervention from the government related to sectoral energy policies, the implemented energy policies and programs which follow market mechanisms will tend to continue the trend of past conditions and will cause the intensity of energy use to remain high (REF scenario). In the REF

scenario, final energy demand is projected to increase from 178 million TOE in 2023 to 337 million TOE by the end of the projection year 2034, increasing at an average of 6.0% per year.

Meanwhile, if the government provides strong encouragement for policies related to the implementation of more intensive energy saving technology with best available technology (BAT) and energy conservation in all sectors (GREEN scenario), then in the same time span, final energy demand will grow slower at an average of 4.7% per year. Thus, total final energy consumption increases to 295 million TOE in 2034.

The development of final energy demand from 2010 to 2034 for the REF and GREEN scenarios is shown in Figure 2.1.

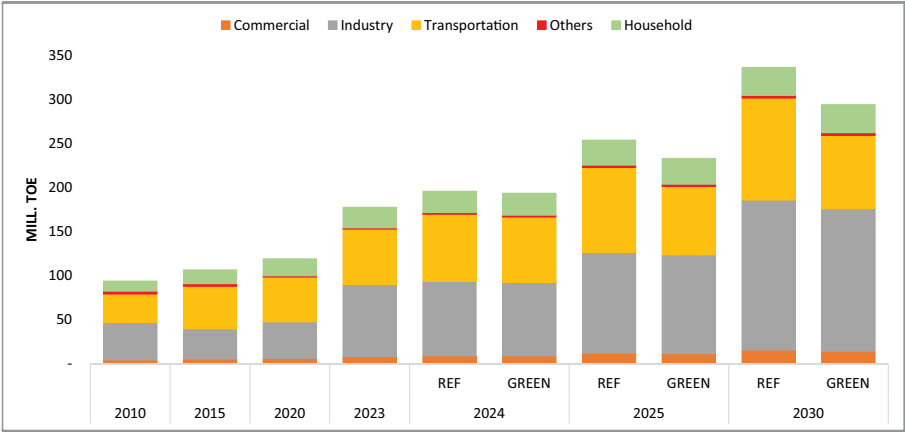


Figure 2.1 Final Energy Consumption by Sector

A significant difference in growth between scenarios occurs in the transportation sector. In the REF scenario, the increase in final energy demand in the transportation sector is on average of 5.7% per year, but in the GREEN scenario, the increase is only 2.6% per year. This is due especially by the use of electric vehicles because electric vehicles can specifically save energy up to 70% to 80%.

2.1.2 Final Energy Consumption by Energy Type

Until now, final energy consumption is still dominated by fossil-based energy. Final energy consumption in 2023 will mostly (75%) be fulfilled by fossil energy. Of the final energy consumption, the contribution comes from oil at 38% (68 million TOE), coal at 25% (44 million TOE) and gas at 12% (21 million TOE). Electricity and NRE consumption contribute at 15% (27 million TOE) and 9% (12 million TOE), respectively.

For the next 10 years, oil will still dominate final energy consumption in Indonesia for both scenarios, reaching 43% in the REF scenario and 31% in the GREEN scenario. However, the growth of oil consumption in the GREEN scenario shows below the average growth of total final energy consumption, which is only 2.6% per year. This condition is influenced by efforts to substitute oil to electric vehicles and increase use of biodiesel.

Coal consumption in both scenarios still shows an increase of 5.6% per year (REF) and 5.0% per year (GREEN) respectively. Likewise, NRE consumption continues to increase at an average growth of 8.2% per year (REF) and 9.4% per year (GREEN) respectively. The increase of NRE in the GREEN scenario is mainly driven by a significant increase in biofuel of around 9.1% per year due to an increase in the FAME (Fatty Acid Methyl Ester) mixture in diesel oil by 40% and ethanol in gasoline by 5% after 2030.

Gas consumption in the next 10 years is projected to grow by around 1.5% per year (REF) and 0.5% per year (GREEN) but the share of gas in total final energy consumption in both scenarios remains stable at around 7% to 8% per year. The development of final energy consumption per energy type and the composition of energy types in 2034 is shown in Figure 2.2.

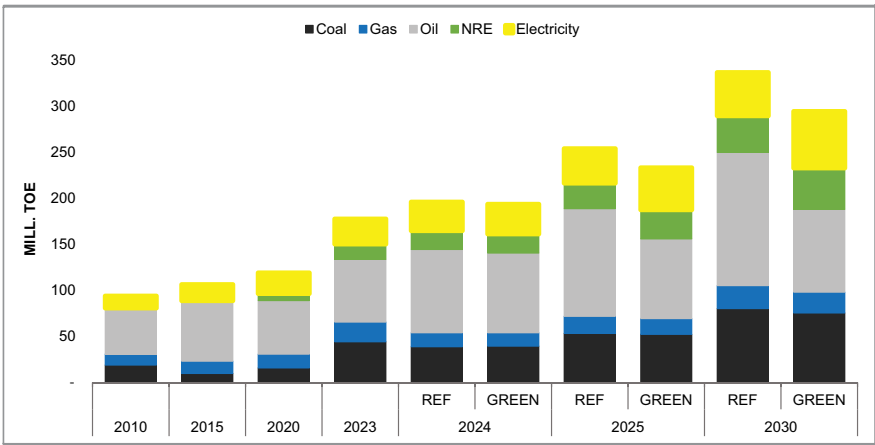


Figure 2.2 Final Energy Consumption by Energy Type

2.1.3 Final Energy Consumption by Region

The Java-Bali region dominates final energy consumption with a share of around 61% in 2023 and 54% in 2034. This is in accordance with the GRDP in the Java-Bali region which reaches 58% of the total national GDP and the population in Java and Bali is around 42% in 2024 (BPS, 2024). With the declining share, the average

growth in energy consumption in the Java-Bali region is the smallest compared to other regions. In the REF scenario, the average national consumption growth is 6.0% per year, while the average consumption growth in Java-Bali is only 4.7% per year (REF) and 3.4% per year (GREEN).

Maluku region final energy consumption is small, but its growth is the fastest, reaching 9.7% per year (REF) and 14.9% per year (GREEN). The rapid growth in energy demand in Maluku is partly influenced by the growing mineral processing and refining industry as an impact of government policies that mandated mining companies to process minerals before exporting their products to the international market.

Similarly, Sulawesi also experienced an average growth in final energy consumption of 9.3% per year (REF) and 10.7% per year (GREEN), mainly influenced by the presence of smelter industry in the Sulawesi region except North Sulawesi and Gorontalo. The overview of final energy consumption by region is as shown in Figure 2.3 below.

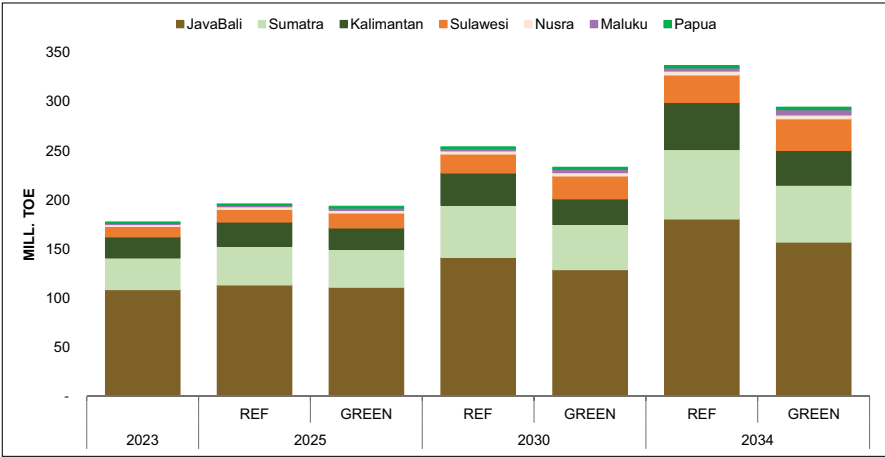


Figure 2.3 Final Energy Consumption by Region

2.2 ELECTRICITY

2.2.1. National Electricity Consumption

Total electricity demand is projected to increase from 315 TWh in 2023 to 547 TWh (REF) and 713 TWh (GREEN) in 2034 or experiencing an average growth of around 5.2% per year (REF) and 7.7% per year (GREEN). In 2023, the share of electricity consumption in the household sector was 39%, but in 2034 it will drop to 26%

(REF) and 32% (GREEN). Meanwhile, the industrial sector, whose share was only 37% in 2023, will dominate energy demand in 2034 to 47% in both scenarios. This is due to the significant industrial growth in Indonesia, which has a direct impact on the significant increase in electricity demand.

In the next 10 years, the transportation sector will experience a very significant growth, namely 22.7% per year (REF) and 36.9% per year (GREEN). However, the demand for electricity is very small compared to other sectors, namely only 1.3 TWh (REF) and 4.3 TWh (GREEN). The rapid growth in electricity consumption is driven by the government's program related to electrification in the transportation sector, especially electric cars and motorcycles. This is influenced by the strong encouragement from the government through the issuance of Government Regulation Number 55 of 2019 on the Acceleration of the Battery-Based Electric Vehicle Program for road transportation which is followed by other supporting regulations such as Presidential Instruction Number 7 of 2022 on the Use of Battery-Based Electric Motor Vehicles as Operational Official Vehicles and/or Personal Vehicles for Central Government Agencies and Regional Governments, Regulation of the Minister of Industry Number 6 of 2023 which regulates a subsidy of IDR. 7 million for the purchase of two-wheeled electric vehicles with a quota of 200,000 in 2023 and 600,000 in 2024. The development of sectoral electricity consumption during the projection year can be seen in Figure 2.4.

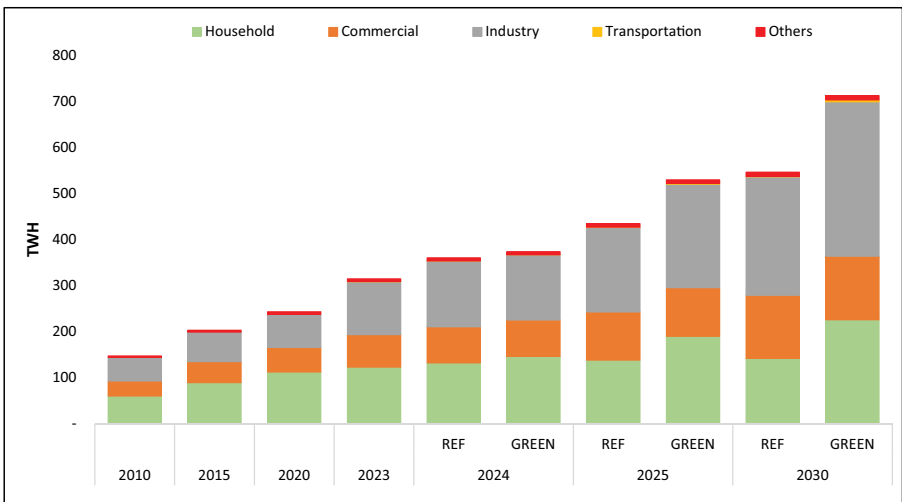


Figure 2.4 National Electricity Consumption by Sector

2.2.2. Electricity Consumption by Region

The largest electricity consumption in Indonesia in 2023 is still in the Java-Bali region, which accounts for 66% of the total electricity consumption. Furthermore, the share is 15% in Sumatra, 11% in Sulawesi, 4% in Kalimantan, 1.9% in Maluku, 1.4% in Nusra and around 0.6% in Papua. The above conditions are in line with the distribution of population and economic mobility in Indonesia which are still concentrated in the Java-Bali region. Papua, of which the island is 6 times larger than Java, only consumes 0.6% of the total national electricity due to the low population and slower economic development in this easternmost region of Indonesia.

For the next 10 years, both in the REF and GREEN scenarios, the demand for electricity will still occur in the Java-Bali region as a result of the high population, economic activity and industrial development which is still concentrated in Java-Bali region. For the GREEN scenario, although the demand for electricity in Java-Bali is still high, its share in 2034 will fall to 56%. On the other hand, the share of electricity consumption in several regions outside Java-Bali will increase, namely from 15% (2023) to 19% (2034) in Sumatra, 4% (2023) to 7% (2034) in Kalimantan and 11% (2023) to 14% (2034) in Sulawesi.

In the GREEN scenario, Kalimantan region experiences a significant increase in electricity demand, which is on average 13.3% per year in line with the development plan for the Indonesian Capital City (IKN) which will become a new center for population and economic growth. Meanwhile, regions outside Java experience an average growth of around 9% per year as an impact of the growth in electricity consumption in the industrial sector outside Java. A complete picture of regional electricity consumption can be seen in Figure 2.5.

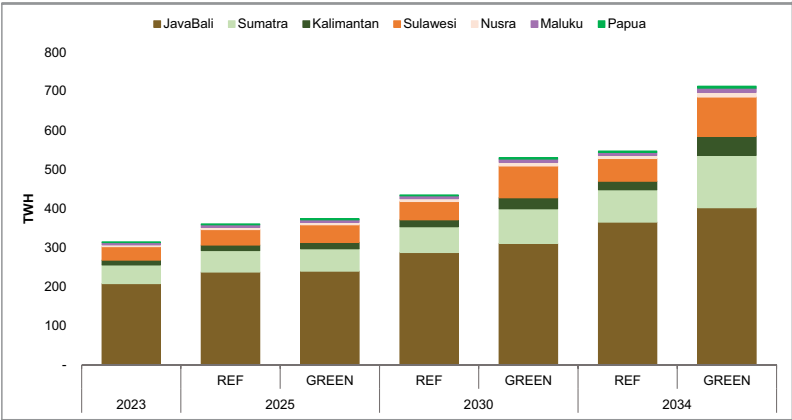


Figure 2.5 Regional Electricity Consumption

2.2.3. National Electricity Production

In line with the growth in electricity demand during the period 2010-2023, electricity production has increased by 5.9% per year or from 170 TWh (2010) to 359 kWh (2023). During this period, coal-fired power plants dominates the electricity production in Indonesia with a growth rate of 8.7% per year or increased from 68 TWh in 2010 to 202 TWh in 2023. In contrast, the electricity production from oil-based power plants decreased from 19 TWh (2010) to 10 TWh (2023) or declined by 4.8% per year. Meanwhile, electricity production from renewable energy-based power plants has increased quite significantly, around 7.5% per year, from 27 TWh (2010) to 69 TWh (2023).

Taking into account around 10% losses in transmission and distribution, total electricity production in the next 10 years is projected to increase to around 584 TWh (REF) and 712 TWh (GREEN) with the main contribution from fossil energy plants, especially coal. In the REF scenario, the share of electricity production from coal-fired power plants is projected to increase from around 56% in 2023 to 67% in 2034. Meanwhile, in the GREEN scenario, the share is projected to decrease to 44%. The decrease in the share of coal in the GREEN scenario is influenced by the biomass co-firing program in several coal-fired power plants. In the REF scenario, it is assumed that co-firing of coal-fired power plants will remain at 5% during the projection period, while in the GREEN scenario, co-firing of coal-fired power plants is targeted to reach 15% starting in 2030 that will affect the decrease in coal-fired electricity production.

On the other hand, electricity production from renewable energy plants in the next ten years, especially in the GREEN scenario, will experience quite significant growth of 10% per year. Meanwhile, for the REF scenario, production growth from NRE power plants only increases by an average of 3.1% per year. Specifically for the GREEN scenario, power plant development is directed at optimizing NRE power plants which have very large potential in Indonesia, especially solar power plant, hydro power plant, wind power plant, geothermal power plant and biomass-based power plant (including co-firing). In addition, to reduce the use of oil in power plants, future development of power plants in the regions is also directed at using gas. Thus, the share of electricity production from diesel power plant will be greatly reduced in 2034 (only around 1.5% of total electricity production). Electricity production by type of energy for both scenarios can be seen in Figure 2.6.

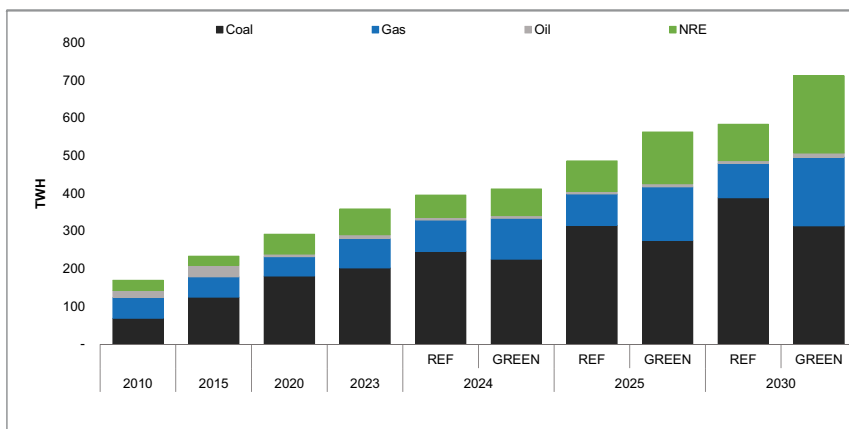


Figure 2.6 Electricity Production by Energy Type

Indonesia has a very large and diverse potential for renewable energy. The development of these sources vary according to the potential and conditions of the region. This provides the advantage of ease in selecting options to meet the requirement for clean electricity in each region. Electricity production from renewable energy plants is projected to increase from 69 TWh in 2023 to 97 TWh (REF) and 205 TWh (GREEN) in 2034. In the REF scenario, electricity production from renewable energy plants in 2034 will grow as it is currently, which is dominated by Hydro Power Plant (36%), Geothermal Power Plant (28%) and Bioenergy Power Plant which include Waste Power Plant, Biomass Power Plant and Biogas Power Plant (28%). Meanwhile, electricity production in the GREEN scenario will be dominated by Solar Power Plant so that its share reaches around 29%. The share of electricity production from Hydro Power Plant is around 25%, while Bioenergy Power Plant and Geothermal Power Plant will contribute 20% and 15% respectively.

The largest increase in NRE electricity production in the GREEN scenario comes from solar power with an average growth of 49.3% per year. Thus, its production increases from 0.7 TWh in 2023 to 59 TWh in 2034. This condition is supported by the continuing decline in the trend of Solar Power Plant installation costs. According to data from IRENA (International Renewable Energy), cost of Solar Power Plant continues to decline from USD 0.445/kWh in 2010 to USD 0.044/kWh in 2023.

In 2032, nuclear power plants will be introduced in the GREEN scenario with electricity production of around 3.2 TWh. Despite that the regulation is not existed, nuclear power plants will be built to meet the need for clean energy. Several locations for nuclear power plant construction projects have been prepared, including in Bangka Belitung Province and West Kalimantan Province. A evolution of the electricity production from NRE power plant can be seen in Figure 2.7.

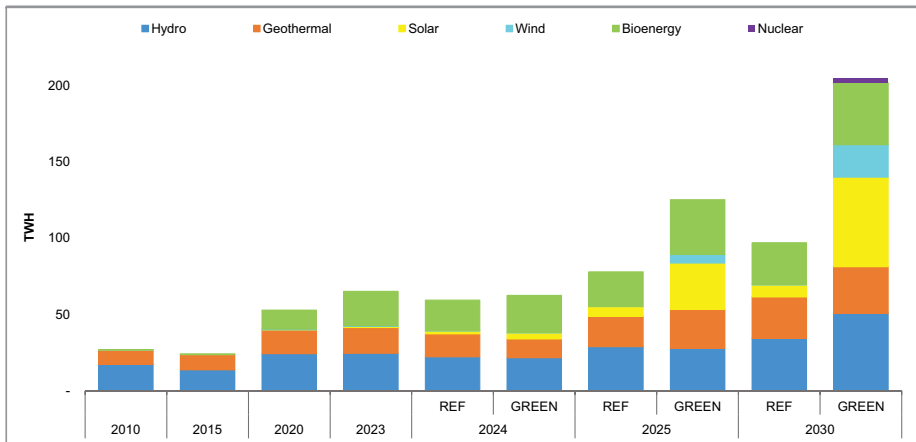


Figure 2.7 NRE Power Plant Electricity Production

2.2.4. Regional Electricity Production

As with the projection of electricity demand by region, the trend of regional electricity production development is similar, which is dominated by the Java-Bali region with a share of 69% (REF) and 59% (GREEN) in 2034. As with electricity consumption by region, in 2034 it is projected that the GREEN scenario electricity production in Java and Bali will decrease. However, outside Java and Bali the share of electricity production will actually increase. A complete picture of regional electricity production can be seen in Figure 2.8.

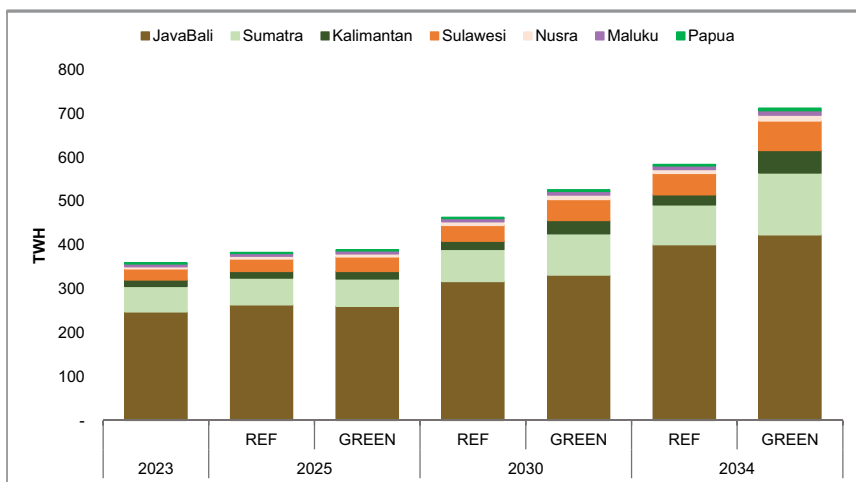


Figure 2.8 Regional Electricity Production

Electricity production from renewable energy-based power plants increased from 69 TWh in 2023 to 97 TWh (REF) and 205 TWh (GREEN) in 2034. If we look at the growth, electricity production from renewable energy power plants is projected to increase significantly in the GREEN scenario in Nusa Tenggara, Maluku and Papua, respectively around 31% per year, 24% per year and 22% per year. This is influenced by the current electricity conditions in eastern Indonesia, where electricity production is still dominated by fossil fuel power plants, especially diesel power plants. The production of electricity from NRE power plants in the REF and GREEN scenarios can be seen in Figure 2.9 below.

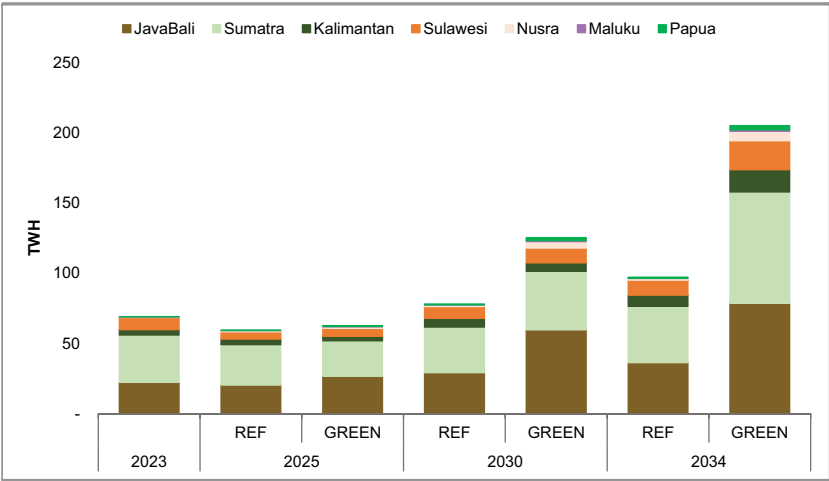


Figure 2.9 Regional NRE Power Plant Electricity Production

2.2.5. National Power Plant Capacity

To meet the electricity demand of the community with sufficient quantity, good quality, and reasonable price, in 2015 the Indonesian government launched the 35 thousand MW Power Plant Development program as one of the government’s strategic projects. The demand for 35,000 MW is an estimate of the growth of electricity that is projected for an average economic growth of 6%. The program has had an impact on increasing the power plant capacity by around 7.9% during the period of 2010-2023, or increasing from 34 GW (2010) to 91 GW (2023). In the last 13 years, coal-fired power plants have still dominated electricity production in Indonesia, increasing from 13 GW (2010) to 50 GW (2023). Meanwhile, oil-based power plants have relatively stable capacity of around 5 GW. Gas-fired power plants have increased slightly from 12 GW (2010) to 23 GW (2023) and NRE power plants have experienced a significant increase from 5 GW (2010) to 13 GW (2023).

Power plant capacity is projected to increase from 91 GW in 2023 to 140 GW (REF) and 213 GW (GREEN) in 2034. All types of power plants experience positive growth, except for diesel power plants which will decline at a rate of 2.5% (REF) and 41.5% (GREEN) per year. This is in line with the dedieselization program which aims to reduce the 5,200 diesel power plants currently still in operation.

The NRE power plants experience the highest growth compared to other energy sources. Thus, NRE power plant capacity will increase from 13 GW in 2023 to 27 GW (REF) and 82 GW (GREEN) in 2034. The development of power plant capacity by scenario can be seen in Figure 2.10.

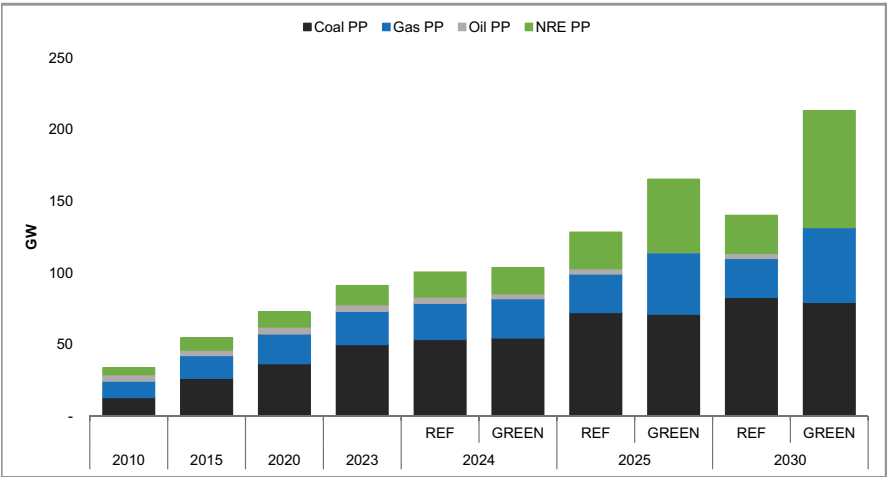


Figure 2.10 Power Plant Capacity by Scenario

The share of Solar Power Plant capacity in the next 10 years will grow significantly. The share of total NRE power plant capacity will increase from 3% (0.4 GW) in 2023 to 19% (5 GW) in the REF scenario and 46% (38 GW) in the GREEN scenario in 2034. The development of power plant capacity in the REF and GREEN scenarios can be seen in Figure 2.11.

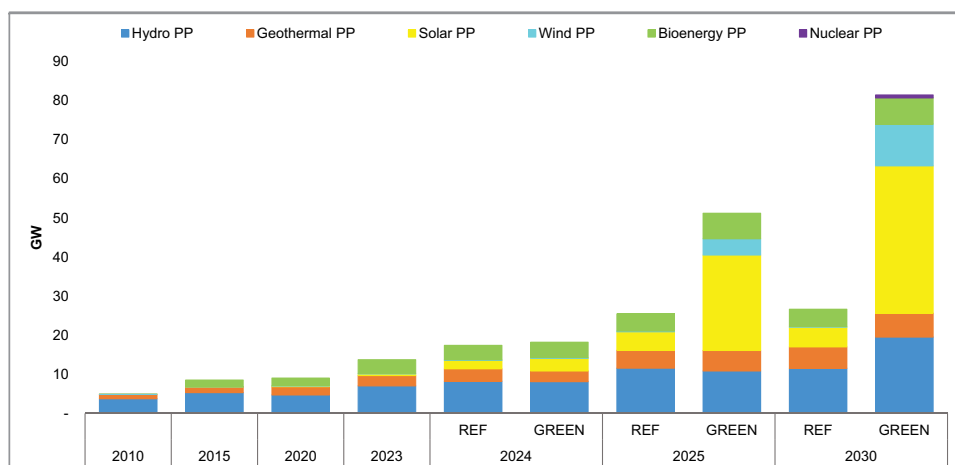


Figure 2.11 NRE Power Plant Capacity Based on Power Plant Type

2.2.6. Regional Power Plant Capacity

In the next 10 years, the population and economic growth centers are projected to still be centered in Java. So in the REF scenario, economic growth in Java-Bali is still high which has an impact on electricity supply. However, for the GREEN scenario, there will be a reduction due to regional development such as the IKN, and the even distribution of industrial economic centers outside Java-Bali. Until 2034, the power plant capacity is still dominant in the Java-Bali region around 58% (REF scenario). While for the GREEN scenario, the contribution of power plant capacity in Java-Bali in 2034 will decrease to 40.5%.

In the GREEN scenario, it is assumed that there will be optimization of the development of NRE power plant, especially the IKN which is fully expected to become a green economy. This has an impact on the high growth of power plant capacity in the Kalimantan region of around 14%. In accordance with the development program of IKN (Indonesian Capital City) and the growing industry in Kalimantan region, the share of power plant capacity will increase from 5% in 2023 to 14% in 2034. The regional power plant capacity can be seen in Figure 2.12.

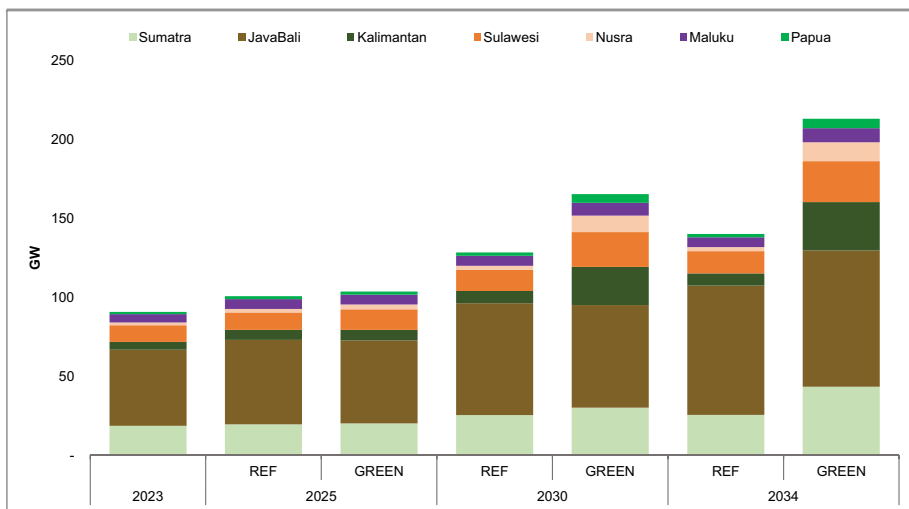


Figure 2.12 Regional Power Plant Capacity

The largest share of NRE power plant capacity in 2023 will be from Sumatra region (44%), followed by Java (35%). This situation will continue in the next 10 years for both scenarios. Specifically in the Maluku region, there will be significant growth in NRE plant capacity of around 42% per year (REF) and 48% per year (GREEN) in the next 10 years. Although the growth is high, the generating capacity only increases from 0.01 GW in 2023 to 0.6 GW (REF) and 1 GW (GREEN) in 2034. The NRE power plant capacity by region can be seen in Figure 2.13.

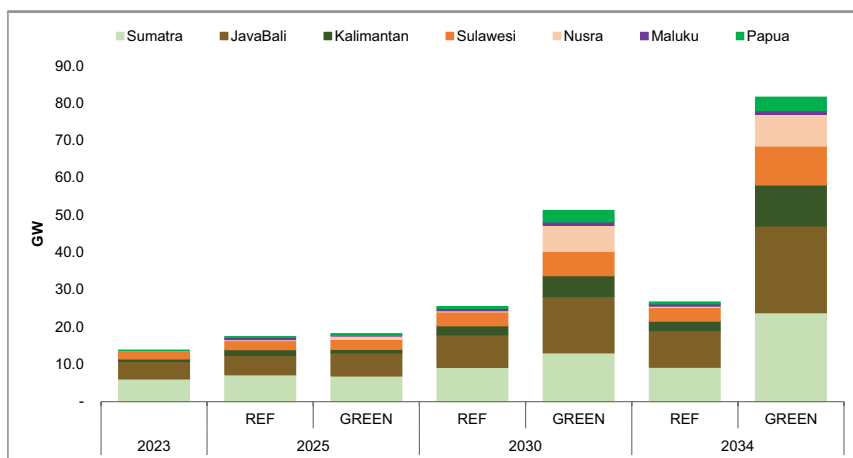


Figure 2.13 Regional NRE Power Plant Capacity

The power plant capacity by region in the GREEN Scenario shows a significant increase in the share of NRE, especially in areas with high renewable energy potential but low total installed capacity, such as Nusa Tenggara and Papua of which NRE power plant capacity has jumped significantly from 8% and 5% in 2023 to 71% and 59% in 2034. In contrast to the conditions in Java and Maluku where the total installed capacity is currently large and dominated by Coal at 59% and 88% respectively, the addition of new NRE capacity only increases the share of NRE capacity from 10% and 0.2% in 2023 to 27% and 11% in 2034 in the GREEN scenario.

In the reference scenario (REF), the growth of NRE power plant capacity is more moderate while maintaining the dominance of fossil energy in several regions. The development of the share of power plant capacity by region is shown in Figure 2.14.

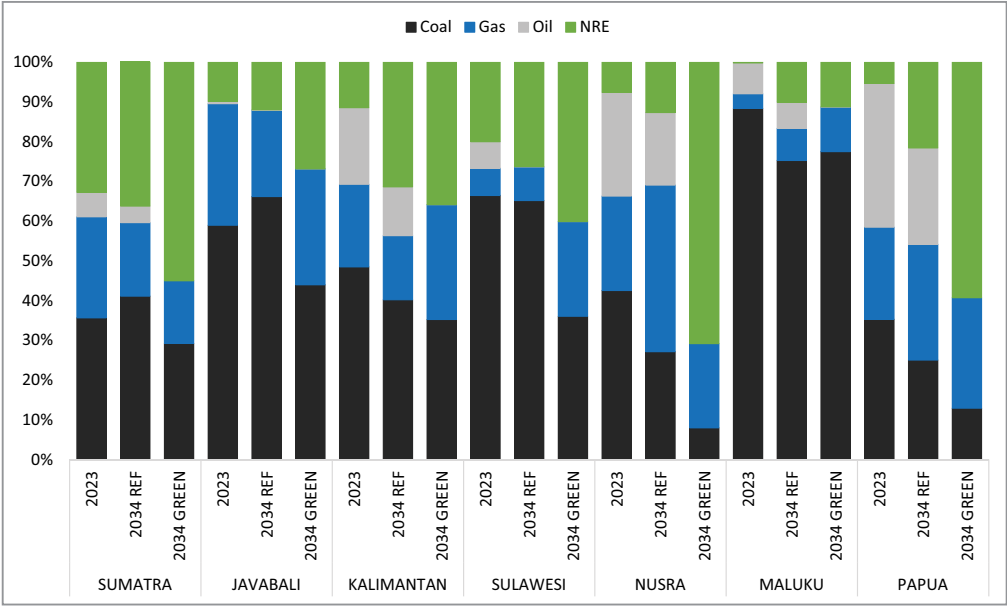


Figure 2.14 Share of Power Plant Capacity by Region

2.3 PRIMARY ENERGY SUPPLY

2.3.1 Primary Energy Supply by Type

Primary energy supply is divided into fossil and non-fossil energy types. Fossil primary energy consists of coal, gas, and oil. Meanwhile, non-fossil primary energy includes various types of new and renewable energy (NRE), namely hydro, solar, wind, geothermal, bioenergy, nuclear energy, hydrogen, and other renewable energies.

Similar to the development of final energy consumption, the primary energy supply has the same trend. During the period 2010-2023, the primary energy supply has increased by 4.3% per year from 151 million TOE (2010) to 259 million TOE (2023). Fossil fuels is still very dominant, only shifting from 43% of oil and 26% of coal in 2010 to 40% of coal and 30% of oil in 2023. Similar to oil supply, gas supply has also decreased from 25% (2010) to 17% (2023). The growth in NRE supply, however, is quite significant and shows the the largest among supplies from other energy sources, namely 11.7% per year. It exceeds coal supply which grows around 7.6% per year. The high growth in renewable energy supply leads to sharp increase of its share in the total primary energy supply, from 5.4% (2010) to 13.3% (2023).

In the future, the total primary energy supply in 2034 will increase by around 5.3% per year (456 million TOE) in the REF scenario and 4.7% per year (429 million TOE) in the GREEN scenario. In more detail, in the REF scenario the primary energy supply of coal grows by an average of 5.8% per year, but in the GREEN scenario it grows slower at 4.1% per year. The decline in coal use in the GREEN scenario is due to the co-firing program in several coal-fired power plants whose mixture continues to increase from the current 5% to 15% starting in 2030. Like past trends, the use of new and renewable energy in the future will continue to be encouraged and even prioritized. Thus, the supply of primary energy from NRE will grow the highest by around 6.8% (REF) and 11.3% (GREEN). This is particularly influenced by the increase in the capacity of NRE power plant, especially Solar Power Plant and the program to increase the use of biofuels. The projection of primary energy supply by type for both scenarios can be seen in Figure 2.15.

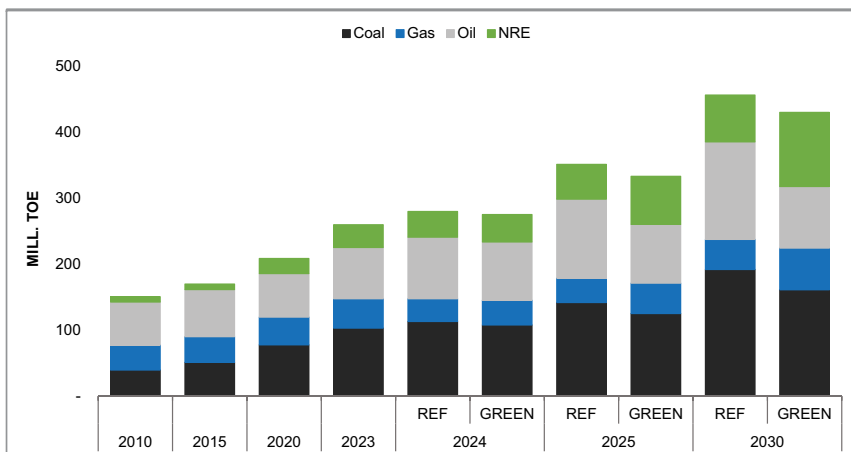


Figure 2.15 Primary Energy Supply Projection by Energy Type

In 2034, NRE supply in REF scenario will be 71 million TOE, or around 15.5% of the total primary energy supply. Meanwhile, in GREEN scenario, NRE supply will be around 112 million TOE, or 26.1% of the total primary energy supply. The overview of the primary energy mix in 2034 for both scenarios can be seen in Figure 2.16.

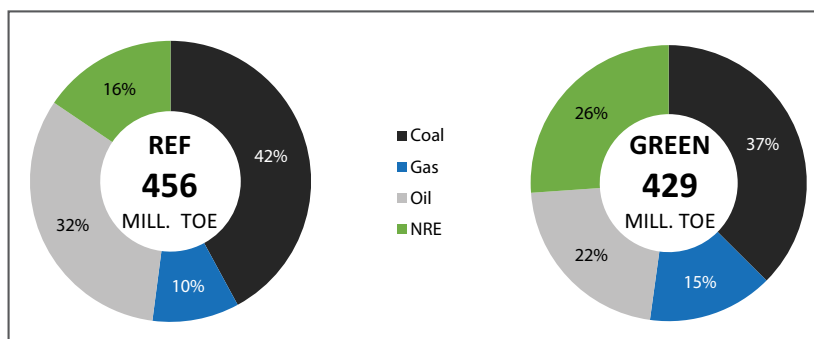


Figure 2.16 Primary Energy Mix Year 2034

2.3.2 Primary Energy Supply by Region

Similar to final energy consumption, Java-Bali is also projected to dominate the primary energy supply, for both the REF and GREEN scenarios. In both scenarios, in 2034 the primary energy supply in the Java-Bali shows a share of 57% (REF) and 55% (GREEN). The shares in Sumatra and Kalimantan are 20% and 12% (REF) and 21% and 11% (GREEN) respectively. The share in other regions is still below 10%. A complete overview of the share of primary energy supply by region can be seen in Figure 2.17.

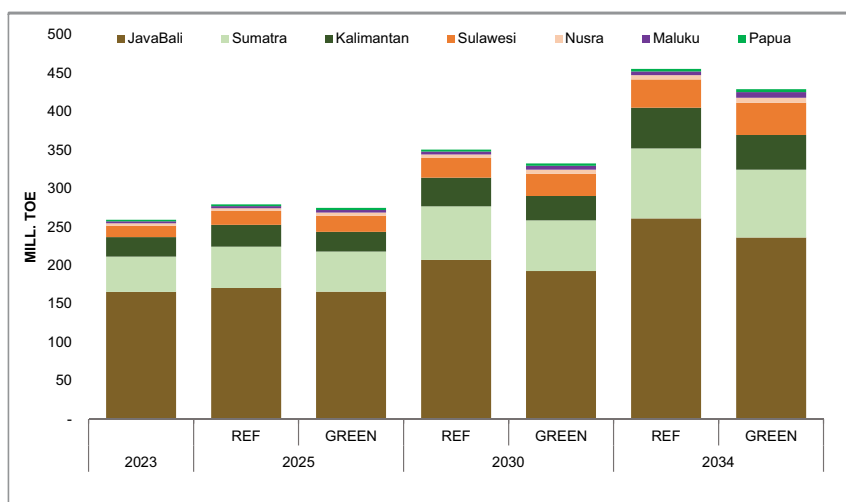


Figure 2.17 Primary Energy Supply by Region

More detailed developments in primary energy supply in each region are shown in Figure 2.18. As shown, there is a shift in primary energy supply in Indonesia, with a significant increase in the use of NRE in the GREEN scenario in 2034 in various regions.

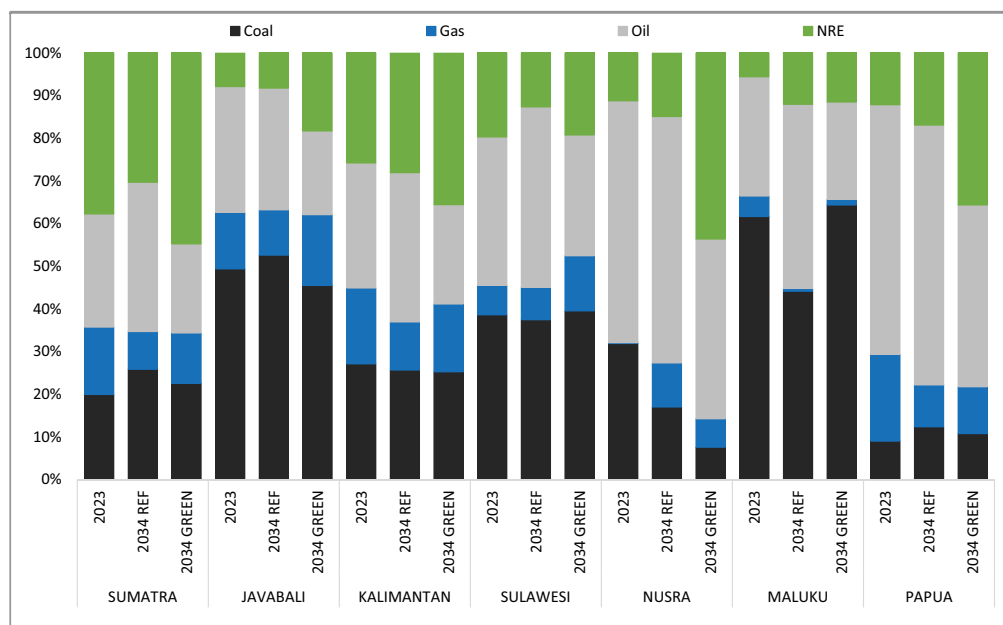


Figure 2.18 Composition of Primary Energy Supply by Region

2.4 CO₂ EMISSION

2.4.1 CO₂ Emission by Sector

The issue on Greenhouse Gas (GHG) emissions in the energy sector cannot be separated from the use of fossil energy since the source of GHG emissions, especially CO₂, is fossil energy. It is certain that when fossil energy is widely used without being accompanied by emission capture or control clean and high efficient technology, the emissions produced will definitely be as high as the energy use.

As stated in the previous sub-chapter, fossil fuels will still dominate energy use in Indonesia. In 2010 to 2023, CO₂ emissions increased from 421 million tons of CO₂Eq (2010) to 752 million tons of CO₂Eq (2023) or increased by an average of 4.6% per year. The industrial and power plant sectors are the largest contributors to emissions with a share of 31% and 30% (2010) and 31% and 38% (2023), respectively.

In 2034, power plants will still be the largest contributor to emissions due to the dominant use of fossil fuels in power plants, especially coal, at around 38% (REF) and 42% (GREEN). Meanwhile, the industrial sector contributes 33% (REF) and 37% (GREEN). The high CO₂ emissions from industry are due to the large amount of equipment in the production, cooling and heating process which still uses fossil fuels and cannot be substituted by electricity.

When viewed based on the GHG emission producing sector in 2034, then the share of 58% (REF) and 56% (GREEN) emissions come from energy use on the demand side and 42% (REF) and 43% (GREEN) of emissions come from the energy supply side, namely from power plants. From the demand side in the REF scenario, the GHG emissions come from the industrial sector (31%), the transportation sector (22%), the household sector (3%) and the commercial sector (1%). Meanwhile, in the GREEN scenario, the industrial, transportation, household and commercial sectors each contribute 36%, 17%, 3% and 0.4%. The development of Greenhouse Gas emissions in the form of CO₂eq from 2010 to 2034 according to the energy user sector can be seen in Figure 2.19.

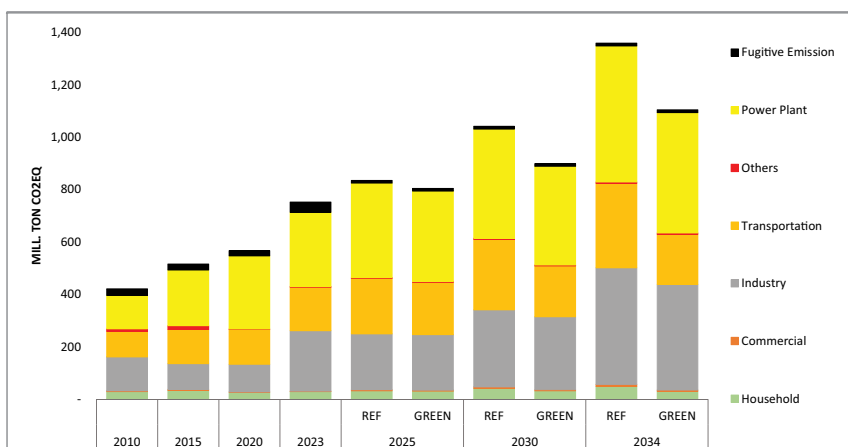


Figure 2.19 CO₂ Emission by Sector

2.4.2 CO₂ Emission by Region

As explained above, the main contributors of GHG emissions are industry and fossil-based power plants. Since industry and power plants, especially fossil-based power plants with large capacities, mostly operate in Java-Bali, the emissions in Java-Bali play as the the main contributors in Indonesia. In 2034, for both scenarios, the Java-Bali region contributes around 64% (REF) and 62% (GREEN) of total national CO₂ emissions. Sumatra contributes between 15% to 16% and for other regions it is below 10% (Figure 2.20).

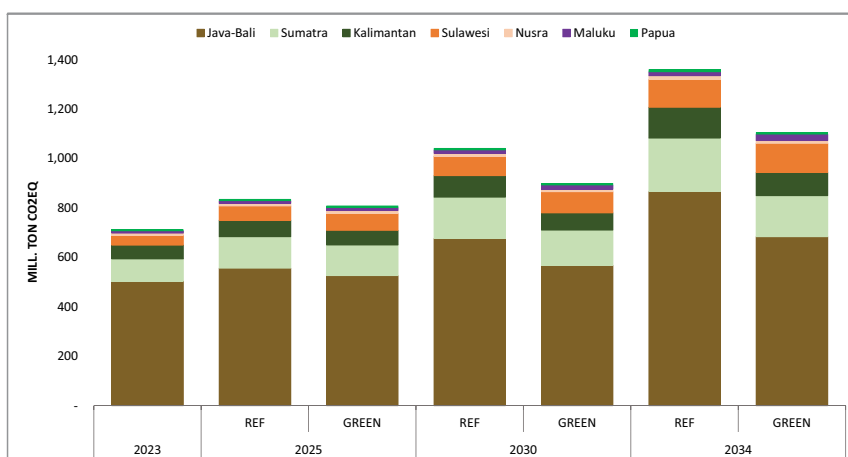


Figure 2.20 CO₂ Emission by Region

2.4.3. Gap Analysis between Emissions and NDC

Total emissions in 2034 are projected to increase to 1,359 million tons of CO₂eq (REF) and 1,105 million tons of CO₂eq (GREEN). In 2030, the emission achievements of both scenarios (1,041 million tons of CO₂eq-REF and 899 million tons of CO₂eq-GREEN) are still lower than the emission target in the energy sector ENDC, which is 1,311 million tons of CO₂eq.

In an effort to keep global temperature rise below 2°C above pre-industrial levels and limit temperature rise to 1.5°C as well as reduce the negative impacts of climate change, Indonesia has committed to reduce Greenhouse Gas (GHG) emissions. The commitment is stated in the NDC Document as a follow-up to the Paris Agreement under the United Nation Framework Convention on Climate Change (UNFCCC) which was adopted at the COP21 Climate Change Conference in Paris on December 12, 2015. The Paris Agreement to The United Nations Framework Convention on Climate Change has been ratified through Law Number 16 of 2016 and the First Nationally Determined Contribution (NDC) has targeted a reduction in emissions of 29% from BAU 2030 with its own efforts and 41% with international assistance. Furthermore, on 23 September 2022, Indonesia submitted the Enhanced NDC to the UNFCCC Secretariat as an increase in the NDC commitment to the Enhanced NDC (ENDC) with a target of 31.89% with its own efforts and up to 43.2% with international support and cooperation.

There are five sectors as the main targets for reducing Greenhouse Gas (GHG) emissions in Indonesia's Enhanced NDC (Nationally Determined Contribution), namely: Forestry and Land Use (FOLU) Sector, Energy Sector, Agriculture Sector, Waste Management Sector, and Industrial Process and Product Use Sector (IPPU).

The projection of total GHG emissions from the five sectors above in 2030 as a reference in BaU level is 2,869 Mton CO₂-eq with a contribution from the energy sector of around 58% or 1,669 Mton CO₂-eq. The emission reduction in energy sector with its own efforts is targeted at 358 Mton CO₂-eq or equivalent to 21.45% of BaU emissions in the energy sector or 12.48% of total BaU emissions. Meanwhile, with international support and cooperation, the emission reduction is targeted at 446 Mton CO₂-eq or 26.72% of BaU emissions in the energy sector or 15.55% of total BaU emissions. The 2030 Greenhouse Gas Emission Reduction Target in the ENDC can be seen in Table 2.1.

Table 2.1 Green House Gas Emission Reduction Target 2030 in ENDC

No.	Sector	2010 GHG Emissions (Juta Ton CO ₂ e)	2030 GHG Emissions (Mill. Ton CO ₂ e)			Emission Reduction					
			BaU	CM1	CM2	CM1	Against the Sector	Against Total	CM2	Against the Sector	Against Total
1	Energy	453	1,669	1,311	1,223	358	21.45%	12.48%	446	26.72%	15.55%
2	Waste	88	296	256	253	40	13.51%	1.39%	44	14.70%	1.52%
3	IPPU	36	70	6	61	7	10.00%	0.24%	9	12.86%	0.31%
4	Agriculture	111	120	110	108	10	8.33%	0.35%	12	10.00%	0.42%
5	Forestry	647	714	217	-15	500	70.03%	17.43%	729	-2.10%	25.41%
Total		1,334	2,869	1,953	1,632	915	31.89%	31.89%	1,240	43.20%	43.20%

The projection results show that the GHG emissions in 2030 reach 1,089 million tons of CO₂eq in the REF scenario and 968 million tons of CO₂eq in the GREEN scenario. These indicate that the REF scenario can achieve a reduction of 34.75% in the energy sector GHG emission target in the ENDC and 20.22% in the total GHG emission target in the ENDC in 2030. Meanwhile, for the GREEN scenario, the achievements are even greater, respectively 42% and 20.22%. The comparison between the reduction in GHG emissions from the 2024 IEO and the energy sector GHG emission reduction target is shown in Table 2.2.

Table 2.2 The Comparison between GHG Emission Reduction Target in REF and GREEN Scenario and ENDC Target 2030 in Energy Sector

	Scenario	2030 GHG Emissions (Mill. Ton CO ₂ e)		Emission Reduction		
		BaU	CM1	CM1	Against the Sector	Against Total
ENDC		1.669	1.311	358	21,45%	12,48%
OEI 2024	REF		1.089	580	34,75%	20,22%
	GREEN		968	701	42,00%	24,43%

2.5 ENERGY INDICATOR

2.5.1 Electricity Consumption per Capita

Indonesia is a country with low electricity consumption per capita. Currently, Indonesia's average electricity consumption is 1,337 kWh per capita. Compared to the ASEAN countries, Indonesia is still far below it or around a third, in which the

average electricity consumption in ASEAN is 3,896 kWh per capita. In 2025, the electricity demand per capita in both scenarios (REF & GREEN) is 1,369 kWh/capita and 1,423 kWh/capita, respectively. These projection results show that electricity per capita is still far from the target in the National Energy Policy, which is 2,500 kWh/capita in 2025. In 2034, the electricity demand per capita will reach around 2,570 kWh/capita in the GREEN scenario and only 1,929 kWh/capita in the REF scenario. The low achievement of electricity per capita is partly due to the average income of Indonesian people which is still relatively low. Thus, the intensive use of electricity is more limited. Despite of the relatively cheap price of electricity, it remains a burden for some people. In addition, it is also influenced by basic data on electricity consumption for own use (captive power) which has not been collected properly, for example in the industrial, mining and other sectors. The projection of electricity consumption per capita for each scenario can be seen in Figure 2.21.

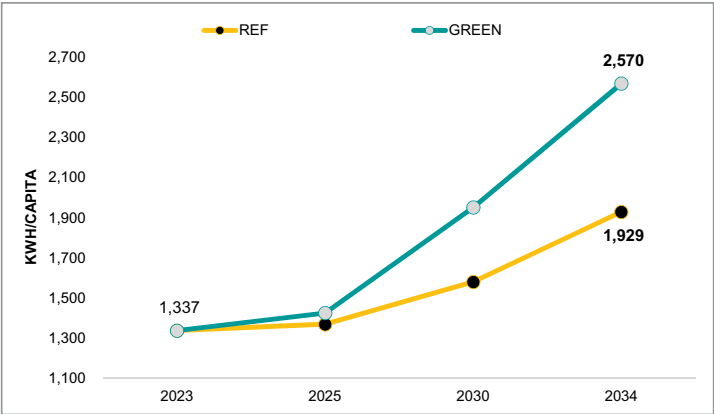


Figure 2.21 Electricity Consumption per Capita

2.5.2 Final Energy per Capita

In 2023, Indonesia's final energy demand per capita is 0.64 TOE/capita. In 2034, it is projected to increase to 1.11 TOE/capita (REF) and 1.00 TOE/capita (GREEN). This figure is lower compared to the global final energy per capita in 2022, which has reached 1.89 TOE/capita (EIA, 2024)

Final energy consumption per capita in the GREEN scenario is lower than in the REF scenario. This is due to lower growth in final energy consumption in the GREEN scenario compared to the REF scenario as a result of implementing energy conservation programs and the use of more efficient equipment in the GREEN scenario, such as the shifting from Internal Combustion Engine (ICE) vehicles to more efficient electric vehicles and the use of energy-efficient equipment in the

commercial and household sectors, including air conditioners, refrigerators and LED lights. The final energy per capita for each scenario can be seen in Figure 2.22.

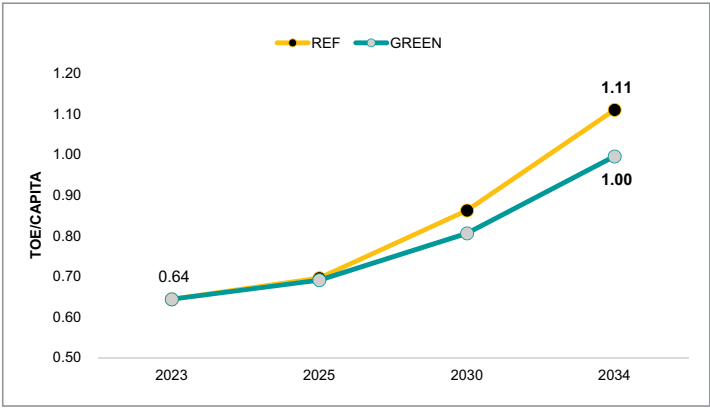


Figure 2.22 Final Energy per Capita

2.5.3 Primary Energy per Capita

In 2034, the primary energy supply per capita of Indonesia will reach 1.50 TOE/Capita in the REF scenario and around 1.45 TOE/Capita in the GREEN scenario. It is lower than the world condition in 2020 which reached 1.7 TOE/Capita (IEA, 2021). The projection of the national primary energy supply per capita for each scenario can be seen in Figure 2.23.

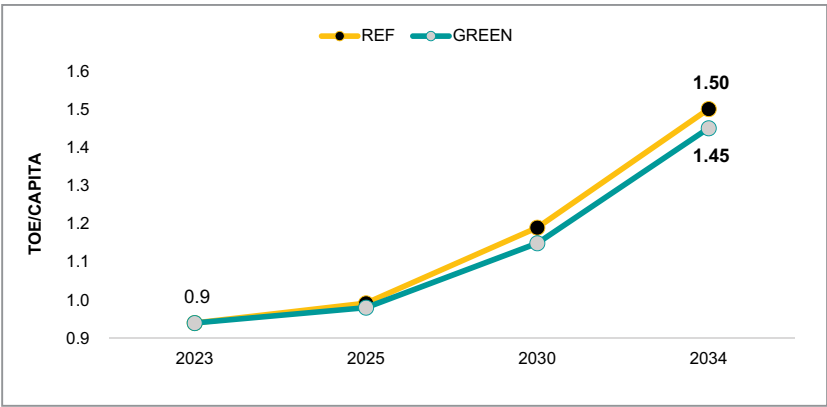


Figure 2.23 Primary Energy per Capita

2.5.4 Emission per Capita

CO₂ emissions per capita in 2023 increase from 2.6 million tons CO₂eq/capita to 4.5 million tons CO₂eq/capita (REF) and 3.7 million tons CO₂eq/capita (GREEN).

The lower emissions per capita in the GREEN scenario shows the direct impact of the use of renewable energy (which is used more in the GREEN scenario) to lower emissions for each person. The projection of national emissions per capita is shown in Figure 2.24 below.

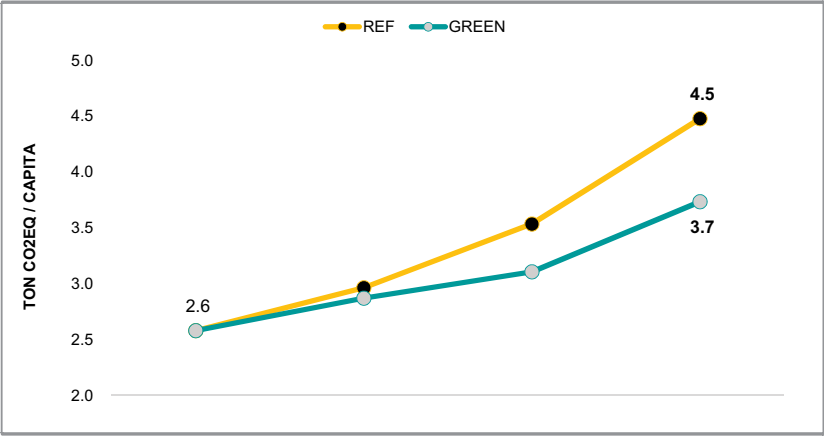


Figure 2.24 CO₂eq Emission per Capita

2.6 PROGRESS OF PRIMARY ENERGY MIX TARGET COMPARED TO RUEN

In the current National Energy Policy (KEN), several national targets have been set, one of which is the energy mix. In the National Energy Policy targets the primary energy mix in 2025 renewable energy will be at least 23%, oil less than 25%, coal at least 30%, and natural gas at least 22%. Furthermore, in more detail, RUEN projects the condition of the primary energy mix each year in order to achieve the target of National Policy Policy. Based on these targets, the modeling results show that the achievement of renewable energy is still quite far from the target. The renewable energy mix in 2025 is estimated to reach 14% for the REF scenario, and 15% for the GREEN scenario. In RUEN, the 2025 target is 30% for coal, 25% for oil and 22% for gas as well as 23% for renewable energy. Of these targets, the share of oil and coal is still higher than the RUEN target for both scenarios. This is because in the RUEN, the target is to limit coal production of 400 million tons starting in 2019, but currently production has increased to around 775 million tons. Meanwhile, the oil share is still far above the RUEN target due to the delayed implementation of the bioethanol program which should have reached 5% starting in 2016. The low achievement of the NRE mix in Indonesia is partly due to the high NRE prices compared to fossil energy. The development of NRE still requires quite high investment. It stil needs further policies and regulations for supporting infrastructure on NRE.

However, the NRE mix is projected to continue increasing with NRE share reaching 23% in 2030 and 26% in 2034. A comparison between primary energy mix projections in 2025 for the REF and GREEN scenarios and RUEN is shown in Figure 2.25. Meanwhile, the NRE target achievement projection is shown in Figure 2.26.

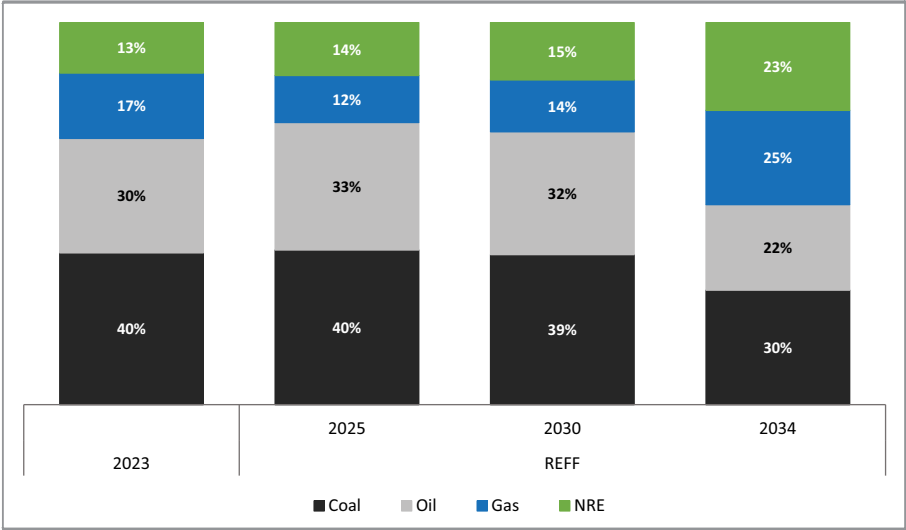


Figure 2.25 Comparison of Primary Energy Mix Projection Year 2025

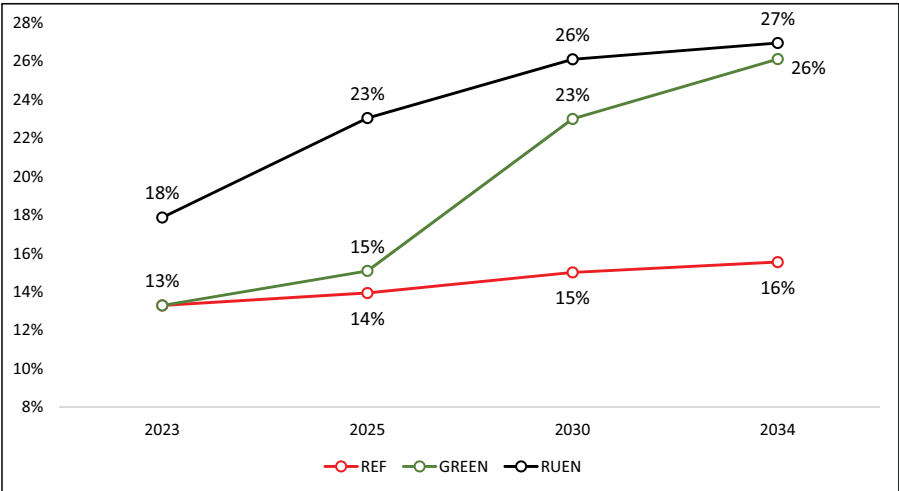
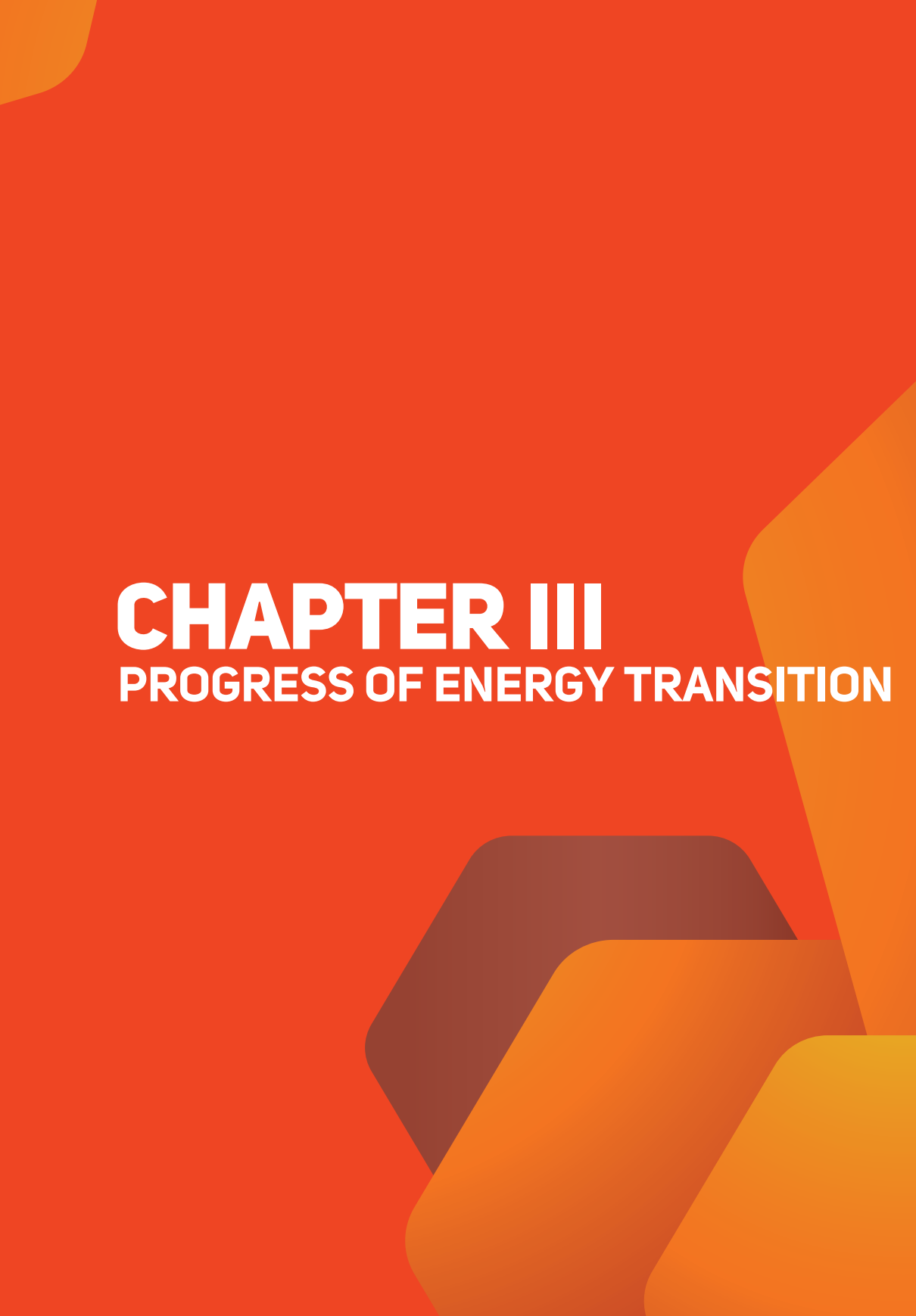


Figure 2.26 NRE Mix Projection Year 2023-2034

The background is a solid orange color. There are several abstract geometric shapes in various shades of orange and brown. In the top left, there's a small orange shape. On the right side, there's a large, light orange shape. At the bottom, there are three overlapping shapes: a dark brown one on the left, a medium orange one in the center, and a light orange one on the right.

CHAPTER III

PROGRESS OF ENERGY TRANSITION



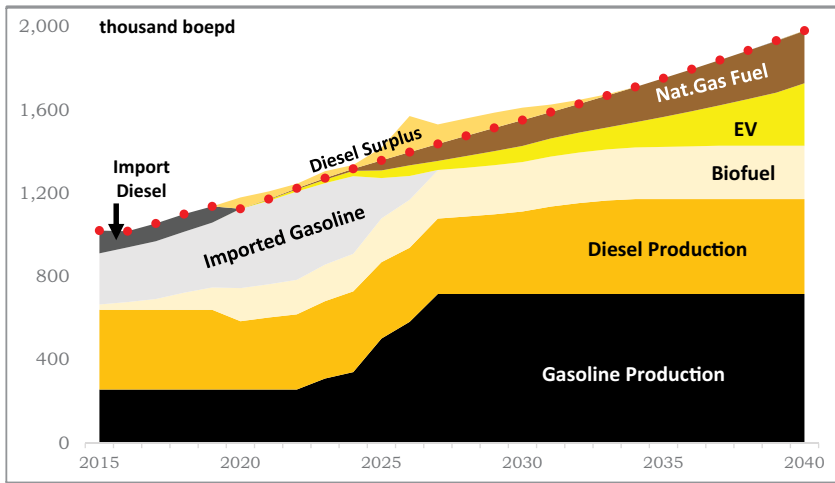
CHAPTER III

PROGRESS OF ENERGY TRANSITION

3.1 STATUS OF ELECTRIC TRANSPORTATION DEVELOPMENT

To encourage the development of electric cars in 2019, the Government issued Presidential Regulation Number 55 of 2019 concerning the Acceleration of the Battery-Based Electric Vehicle (KBLBB) Program for Road Transportation. This regulation was followed by several derivative regulations, including Presidential Instruction Number 7 of 2022 concerning the Use of Battery-Based Electric Motor Vehicles (Battery Electric Vehicles) as Operational Official Vehicles and/or Personal Official Vehicles of Central Government Agencies and Regional Governments. In addition, the Indonesian Government has also issued policies and incentives to support the growth of the electric vehicle market including tax incentives, exemption from import duties for electric vehicle components, and reduction in electric vehicle registration fees. These steps aim to increase the penetration of electric vehicles as an effort to reduce dependence on fuel imports.

One of the programs in the Grand National Energy Strategy (GSEN) is the reduction of fuel imports through increasing CNG vehicles, the use of biofuel, and the use of Battery-Based Electric Vehicle. The target number of electric cars is around 2 million units and electric motorcycles reach 12 million units in 2030. This Battery-Based Electric Vehicle program is expected to stop fuel imports before 2030, as can be seen in Figure 3.1.



Source: GSEN, 2020

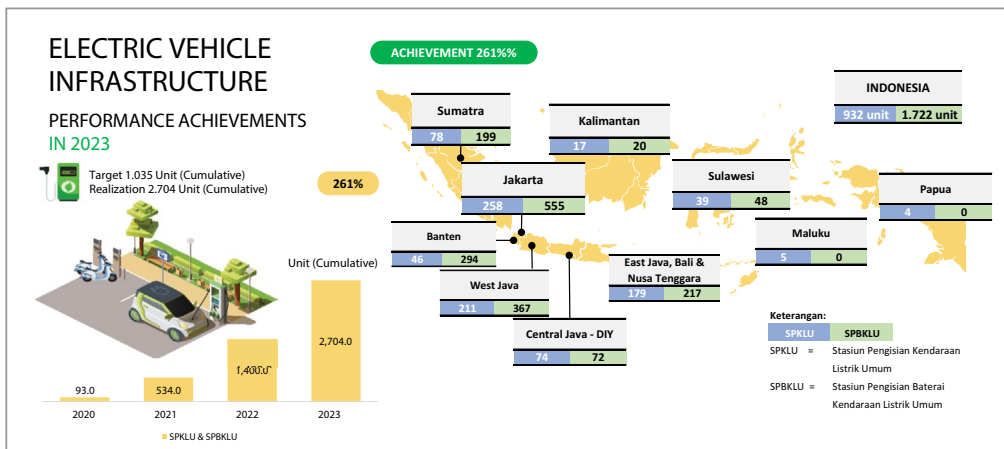
Figure 3.1 GSEN Program for Fuel Import Termination

In early 2024, the total registered electric vehicles reached 133,225 units, consisting of 109,576 Electric Motorcycles, 23,238 Electric Cars, 320 Electric Three-Wheeled Vehicles, 81 Electric Buses, and 10 Electric Commercial Vehicles.

The population of electric vehicles in Indonesia continues to increase every year, especially in the last two years. In 2019, there were only 1,437 electric vehicles in Indonesia. This number jumped to 41,743 units in 2022 and skyrocketed to 133,225 units in early 2024.

This transition becomes a concern for online transportation companies such as GOJEK and GRAB in an effort to reduce their carbon footprint. Both transportation companies have started providing services using electric motorbikes and cars of around 13,500 units in 2023, consisting of 8,500 units from Grab and 5,000 units from Gojek.

For electric vehicle charging stations, both SPKLU and SPBKLU, up to 2023, 2,704 units have been successfully built, far exceeding the target of 1,035 units (261%). The distribution of Public Electric Vehicle Charging Stations (SPKLU) and Public Electric Vehicle Battery Exchange Stations (SPBKLU) in Indonesia in 2023 is shown in Figure 3.2.



Source: Directorate General of Electricity, MEMR

Figure 3.2 Distribution of Public Electric Vehicle Charging Station and Public Electric Vehicle Battery Exchange Station in Indonesia Year 2023

3.2 STATUS OF HYDROGEN DEVELOPMENT

In Indonesia, hydrogen development is still in the research and pilot project stage and there hasn't any commercial projects yet. Research on hydrogen as a fuel has been conducted by the Agency for the Assessment and Application of Technology (BPPT), University of Indonesia (UI), Bandung Institute of Technology (ITB) and Lemigas since 2007. Furthermore, in 2012, ITB created the concept of a fuel cell car and the Korea International Cooperation Agency (KOICA) provided a grant to the Indonesian government for the construction of a 300-kilowatt (kW) fuel cell pilot project in Ancol, Jakarta. Then in 2014, the Indonesian Association for Fuel Cell and Hydrogen Energy (INAFHE) was established to accelerate hydrogen development in Indonesia. In 2017, PT Telkomsel developed a fuel cell at the Base Transceiver Station (BTS) as backup power. After that, in 2018, BPPT and Toshiba ESS signed a cooperation agreement regarding the Development of the H2One Autonomous Hydrogen Energy System for Off Grid Systems. In 2019, the Ministry of Energy and Mineral Resources, PT Kereta Api Indonesia (KAI) and Allstorm began collaborating in the development of hydrogen-fueled trains. PT HDF Energi took the initiative to develop GREEN hydrogen from a hybrid Solar Power Plant and Wind Power Plant in Sumba Island with a capacity of 7-8 MW during the day and 1-2 MW at night from hydrogen storage. In addition, there is also cooperation between PT Pertamina and GIZ in developing a pilot project for GREEN hydrogen from geothermal energy. In 2023, PT PLN (Persero) through the subholding PLN Nusantara Power inaugurated the first GREEN Hydrogen Plant (GHP) in Indonesia located in the Muara Karang Gas Steam Power Plant

(PLTGU) area, Pluit, Jakarta. This GHP is 100 percent sourced from solar energy and is capable of producing 51 tons of hydrogen per year. Of the total production, 43 tons can be used for 147 cars that can travel 100 km every day.

Indonesia is developing five green hydrogen projects to produce clean hydrogen with a total capacity of 1.4 kilo tonnes per annum (ktpa) to utilize the country's renewable energy potential and carbon storage capacity. The projects include the Batam Bintan GREEN Hydrogen Cluster, PT Pertamina's Clean Hydrogen Cluster, Cilegon Clean Hydrogen Cluster, North Sulawesi GREEN Ammonia Cluster, and the Sumatra-Java Blue Ammonia Project, with operational targets between Q1 2027 and Q1 2030. Indonesia is positioned as a regional leader in clean hydrogen production due to its abundant natural gas resources, CO₂ storage capacity, renewable energy potential, and geographical proximity to major hydrogen markets such as Japan, South Korea, and Singapore. This promises significant economic benefits and supports global decarbonization goals.

Geographically, South Korea and China are the main markets for hydrogen vehicles, especially for commercial vehicles. Hyundai Nexo is the leading model, while Toyota Mirai is facing declining sales.

In Indonesia, hydrogen development shows great potential to support the energy transition and achieve decarbonization targets. Hydrogen initiatives are emerging though still in their early stages. In 2023, the main focus was developing ecosystem development and infrastructure pilot project. PT PLN has inaugurated the first hydrogen filling station in Jakarta. In addition, Toyota and Pertamina are collaborating to promote this technology with the aim of supporting Indonesia's carbon neutral target by 2060.

3.3 STATUS OF ELECTRIC STOVE UTILIZATION

The Indonesian government has been encouraging the use of electric stoves as part of its energy transition efforts to reduce dependence on fossil fuels and reduce greenhouse gas emissions. The government has also conducted pilot project on the use of electric stoves in several areas and distributed electric stoves for free to the public. This pilot project aims to measure the effectiveness and public acceptance on the use of electric stoves.

The government plans to stop subsidies for LPG in the long term and replace it with the use of electric stoves. This is carried out to reduce the burden of energy subsidies and encourage the use of clean energy. Meanwhile, one of the

efforts to reduce LPG imports is to substitute it with electric stoves. In the GSEN document, several programs to reduce LPG imports include: Development of City Gas Networks, Increasing Production Capacity from Oil Grass Root Refineries, Development of Dimethyl Ether (DME) and the use of electric stoves.

Since its introduction in 2019, the adoption of electric stoves in Indonesia is still very low. Data shows that only around 0.3% of Indonesian households use electric stoves for cooking in 2023.

The low use of electric stoves in society is due to a number of obstacles that need to be overcome to enable the energy transition to run effectively, such as high initial costs, limited electricity infrastructure, changes in habits, household electric power and dependence on LPG subsidies.

To overcome these obstacles, it requires improvement in electricity infrastructure, subsidies or incentives for purchasing electric stoves, and effective education programs to increase public awareness on the long-term benefits of using electric stoves.

3.4 STATUS OF CCS/CCUS DEVELOPMENT

The energy transition is carried out through the process of transforming the use of fossil fuels using clean technology, accelerating the use of new renewable energy and increasing energy conservation activities. The energy transformation towards Net Zero Emission is expected to meet energy independence, energy security, sustainable development, climate resilience and low carbon conditions. Carbon Capture and Storage (CCS) and Carbon Capture, Utilization, and Storage (CCUS) technologies are developing rapidly in Indonesia, driven by the government's commitment to reduce greenhouse gas (GHG) emissions in accordance with the Paris Agreement.

CCS/CCUS captures and stores carbon dioxide (CO₂) produced from various sources, such as coal-fired power plants, chemical plants, and other industries. By reducing the amount of CO₂ released into the atmosphere, this technology can help reduce greenhouse gas emissions that cause global warming. In addition to storage, CO₂ can be used for other applications, such as as a feedstock for synthetic fuels or chemical products (utilization). The use of CCS/CCUS also plays the role as a bridge in the transition from fossil-based energy to clean energy. By reducing CO₂ emissions from hard to abate sources, this technology can provide additional time to develop and implement more sustainable renewable energy sources.

CCS and CCUS have good prospects in Indonesia as part of the energy transition and carbon emission reduction solutions. However, its successful implementation requires strong commitment from the government, financial support, and cross-sector collaboration. With the increasing demand for clean energy and emission reduction, CCS/CCUS has the potential to become an important part of Indonesia's climate mitigation strategy in the upcoming decades.

In 2020, the Indonesian Government conducted a feasibility study of the CCUS Pilot Project in Gundih Oil and Gas Field, Central Java. The total potential of CO₂ reduction in this pilot project is projected to be 2.92 million tons over 10 years. PT Pertamina also has formulated a roadmap for implementing CCUS in 7 locations to enhance the oil and gas production, namely in Subang, Cilamaya, Jatibarang, Merbau, Jambaran Tiung Biru, Natuna and Matindok. Based on data from the Directorate General of Oil and Gas, there are 16 oil fields that will implement CCS and will be onstream by 2030 at the latest.

Carbon Capture, Storage, and Utilization (CCS/CCUS) technology is developing rapidly in Indonesia and has a very promising prospects in the future. However, it also has quite tough challenges in its development. The following are the latest developments, future prospects and challenges of CCS/CCUS technology in Indonesia:

Current status of CCS/CCUS development:

1. Regulation and Policy:

- a. Indonesia has included CCS/CCUS as part of its climate change mitigation strategy in the Enhanced Nationally Determined Contribution (ENDC) and the Long-Term Strategy for Low Carbon and Climate Resilience (LTS-LCCR) documents.
- b. The Indonesian government has issued several regulations to support the implementation of this technology, including Presidential Regulation No. 98 of 2021 on the economic value of carbon and OJK Regulation No. 14 of 2023 on carbon trading. In addition, the government is preparing special regulations to support the implementation of CCS/CCUS, including its legal aspects, incentives, and operational standards.

2. Pilot Project:

Pertamina and PLN have been pioneers in the development of this technology. Pertamina has carried out various CCS/CCUS development initiatives, such as in Asri Basin in Northern Java as well as in Jatibarang and Sukowati Fields.

- a. Gundih Field (Pertamina): The first CCS project in Indonesia, designed to capture CO₂ from natural gas production.

- b. Sukowati Field: Pertamina is also developing studies for Enhanced Oil Recovery (EOR) based-CCUS which utilizes CO₂ to increase oil production.
- 3. Technology and Infrastructure:
 - a. CCS/CCUS technology is still in the development stage in Indonesia, with a focus on applications in the fossil energy sector, such as coal-fired power plants (PLTU) and heavy industry (cement, steel).
 - b. Geological studies to discover CO₂ storage formations, such as sandstone basins in the Kalimantan and Sumatra regions, are ongoing.
- 4. International Collaboration: Indonesia is also in collaboration with international partners such as Exxon Mobil, JERA Co., Inc., and JGC Corporation to strengthen the capacity for implementing CCS/CCUS technology.

3.5 STATUS OF NUCLEAR DEVELOPMENT AND NEPIO

Currently, Indonesia has three long-operating research nuclear reactors, namely TRIGA 2000 Reactor in Bandung, Kartini Reactor in Yogyakarta, and GA Siwabessy Reactor in Serpong. The government is trying to improve the skills and capability of human resources in the nuclear field to support the development of nuclear power plants.

Furthermore, the development of nuclear power plants in Indonesia will begin on a small scale, with an initial capacity expected to reach 250 MW in 2032. The roadmap for nuclear power plant development has been included in the national energy strategy. The government also plans to increase the capacity of nuclear power plants to 35 GW by 2060.

The existence of nuclear mining materials in Indonesia is spread across 26 locations in Sumatra, Kalimantan, Sulawesi, Maluku, and Papua. The estimate of uranium and thorium resources have been carried out in 9 deposit locations. In 2021, the amount of uranium resources was 89,483 tons of U₃O₈ (the heat generation value for Uranium of this amount is equivalent to the heat generation of 1.85 trillion tons of coal). Meanwhile the amount of thorium resources was 143,234 tons. Details of radioactive mineral resources in Indonesia can be seen in Table 3.1.

Table 3.1 Indonesia Radioactive Mineral Resources Year 2021

No.	Potential Sector	Resources								
		Measured		Indi- cated	Consessed		Hypotethic		Speculative	
		U ₃ O ₈	Th	U ₃ O ₈	U ₃ O ₈	Th	U ₃ O ₈	Th	U ₃ O ₈	Th
1	Aloban, Sibolga - North Sumatera				490*					
2	Singkep - Riau						1,298*	433*		
3	Bangka Belitung	2,840*	4,729*				1,224*	10,361*	25,715*	111,298*
4	Ketapang - West Kalimantan						736*	4,767*		
5	Kalan, Melawai - West Kalimantan	2,394*		5,903*	2,914*		5,058*			
6	Mentawa dan Darab, Seruyan - Central Kalimantan				623*		9,669*			
7	Katingan - Central Kalimantan						572*	2,261*		
8	Kawat, Mahakam Hulu - East Kalimantan						17,861*			
9	Mamuju - West Sulawesi				769*	3,424*	3,023*	3,138*	8,393*	2,823*
Total		5,234*	4,729**	5,903*	4,796*	3,424**	39,441*	20,960**	34,108*	114,121**
		Mea- sured U	Mea- sured Th	Indicat- ed U	Conses- sed U	Conses- sed Th	Hypo- tethic U	Hypo- tethic Th	Specu- lative U	Specu- lative Th
Total U ₃ O ₈		89,483*								
Total Th		143,234**								

Sumber: BRIN, 2021

Note: (*) Satuan Ton U₃O₈

(**) Satuan Ton Th

For countries that do not yet have a nuclear power plant and intend to develop one, it is strongly recommended by the International Atomic Energy Agency (IAEA) to establish a Nuclear Energy Program Implementing Organization (NEPIO) as part of the government's commitment to develop nuclear energy. Therefore, in 2021, the Minister of Energy and Mineral Resources Decision Number 250.K/ HK.02/ MEM/ 2021 concerning the Preparation Team for the Establishment of a Nuclear Energy Program Implementing Organization has been issued. Currently, Nuclear Energy Program Implementation Committee is being proposed to the Ministry of Administrative and Bureaucratic Reform to obtain the draft of Presidential Decree on NEPIO.

3.6 STATUS OF IMPLEMENTATION ON CARBON PRICING

Carbon Pricing is an approach which includes various mechanisms to assign economic value to carbon emissions, whether through emission reductions, carbon trading, or carbon-based incentives. In 2023, the implementation of the Carbon Pricing in Indonesia has shown progress with the start of domestic carbon trading and regulatory framework support. However, challenges in technical aspects, regulations, and industrial sector readiness still need to be overcome. With correct approach, the Carbon Pricing can be an important instrument in Indonesia's transition to a low-carbon economy.

Here are some important points regarding the status and development of the Carbon Pricing in Indonesia:

1. **Regulations and Policies:** The government has issued several important regulations, including Presidential Regulation Number 98 of 2021 concerning the Carbon Pricing and Regulation of the Minister of Environment and Forestry Number 21 of 2022 concerning the Procedures for the Implementation of the Carbon Pricing. These regulations serve as a basis for business sectors to reduce their carbon emissions.
2. **Carbon Trading:** In 2023, Indonesia launched the Indonesia Carbon Exchange (IDX Carbon) which allows formal carbon trading. This is part of the effort to achieve the Net Zero Emission (NZE) target by 2060.
3. **International Commitment:** Indonesia has demonstrated a strong commitment to reduce carbon emissions through the Enhanced Nationally Determined Contribution (ENDC), with a target of reducing emissions in 2030 by 31.89% with its own efforts to 43.2% with international support compared to BaU.
4. **Cooperation and Support:** The government is working with various institutions, including OJK and the Ministry of Environment and Forestry, to ensure that the carbon trading mechanism runs smoothly. In addition, there is cooperation with the private and international sectors to support the implementation of the carbon pricing.
5. **Economic Impact:** The carbon pricing is considered a key milestone in the energy transition and achievement of net zero emission targets. With carbon trading regulations and mechanisms, it is hoped that business entities will be more motivated to invest in environmentally friendly technologies.

In addition to Carbon Pricing, there is also a Carbon Tax. Carbon Tax is a fiscal instrument that imposes a tax on carbon emissions or the use of fossil fuels based on their carbon content. Carbon tax, is also known as carbon emission tax or

carbon pollution tax. Carbon tax is usually imposed on companies or individuals that produce carbon emissions, based on the amount of carbon dioxide or carbon equivalent produced from their activities.

In this case, the government sets a fixed price per ton of carbon (for example, IDR 30,000 per ton of CO₂). The tax is then imposed directly on carbon emissions from specific sources (such as power plants or heavy industry). Indonesia plans to introduce a carbon tax with an initial rate of IDR 30,000/ton of CO₂ in the energy sector, although the implementation is still pending.

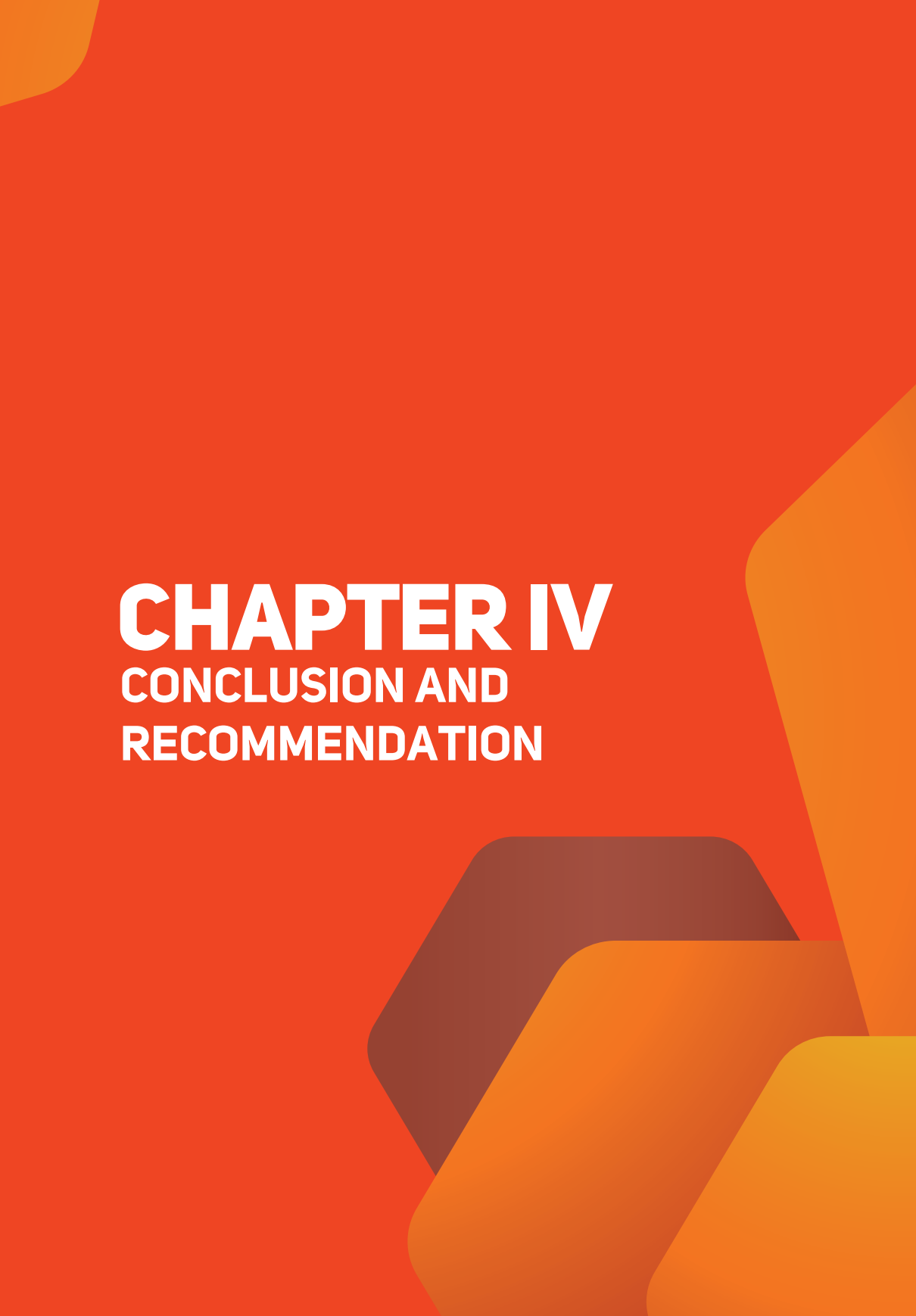
The implementation of carbon tax in Indonesia is regulated in Law Number 7 of 2021 concerning Harmonization of Tax Regulations. This law provides a legal basis for the government to impose carbon tax and regulates the rates and mechanisms for its implementation.

Carbon tax in Indonesia uses a cap and tax scheme, which regulates the permissible emission limits for each industry. Companies that exceed the emission limits will be subject to carbon tax according to the set rates. A carbon market trial was conducted in 2021 for power plants. In the trial, the upper limit value (CAP) for GHG emissions was determined based on the weighted average GHG emission intensity value in 2019 for 3 (three) groups of Coal Fired Power Plant participants in the carbon trading trial and considering the excess allocation of emission quotas. Meanwhile, the distribution of capacity is determined based on the Regulation of the Minister of Energy and Mineral Resources Number 9 of 2020 concerning the Efficiency of Electricity Supply of PT PLN (Persero). The CAP value based on the classification of generating capacity is shown in Table 3.2 below.

Table 3.2 Classification on Capacity and Value of GHG Emission CAP

Type of Power Plant	Installed Capacity (MW)	Cap Value (ton CO ₂ /MWh)
Coal PP	X > 400	0.918
Coal PP	100 ≤ X ≤ 400	1.013
Mine Mouth PP	100 ≤ X ≤ 400	1.094

The implementation of the carbon tax in Indonesia was initially planned to start in 2023. However, the Ministry of Finance has decided to postpone the implementation of the carbon tax until 2025 since it requires technical preparations and information dissemination.

The background is a solid orange color. It features several large, abstract, rounded geometric shapes in various shades of orange and brown. One large brown shape is on the right side, and another is at the bottom right. There are also some smaller orange shapes in the top left and bottom right corners.

CHAPTER IV

CONCLUSION AND RECOMMENDATION



CHAPTER IV

CONCLUSION AND RECOMMENDATION

4.1 CONCLUSION

1. Final energy consumption in 2034 is projected to increase to 337 million TOE (REF Scenario), where the industrial sector and transportation sector are the main consumers with a share of 50% and 34% respectively. Implementing energy conservation and massive application of technological efficiency (GREEN Scenario), the growth of final energy consumption can be suppressed to 295 million TOE.
2. Based on energy type, final energy consumption in the next 10 years will still be dominated by oil (31% - 43%) and coal (21% - 24%), with the largest consumption in the Java-Bali region (54%). The development of the Indonesian Capital City (IKN) has an impact on increasing electricity demand at an average of 13.3% per year in the Kalimantan region under the GREEN Scenario.
3. Electricity consumption is projected to increase from 315 TWh (2023) to 547 TWh (REF) and 713 TWh (GREEN) in 2034. In the next 10 years, there will be a shift in electricity consumption in the household sector with the declining share from 39% (2023) to 26% - 32% (2034). Meanwhile, the substitution of electricity-based equipment in the industrial sector has an impact on the increasing share of electricity consumption of the sector from 37% (2023) to 47% (2034). The electricity consumption in the transportation sector, which currently is below 1%, will experience an increase of around 23% - 37% per year due to the significant growth of electric vehicles..
4. The share of NRE in total primary energy supply will increase from 13.3% (2023) to 26% (GREEN) in 2034 as fossil oil share decreases biofuels utilization

increase, and a more massive NRE power plants development (especially Biomass Cofiring and Solar Power Plant) is being pursued.

4.2 RECOMMENDATION

1. To suppress the growth of energy consumption, the government needs to emphasize energy efficiency policies in each sector. In the household sector, the determination of energy-efficient equipment needs to be expanded by adding other types of equipment labeled energy efficient besides air conditioners, refrigerators and lamps.
2. In an effort to reduce the use of fossil energy, especially coal in power plants and oil in transportation as well as to increase the use of new and renewable energy, related policies are required, which include, among others:
 - Fossil Fuel Subsidy Reduction: reallocating fossil fuel subsidies to renewable energy to increase competitiveness.
 - Accelerating the Implementation of Carbon Pricing: Carbon Trading, Carbon Tax, and Performance-Based Payments in high-emission producing sectors.Specifically for Solar Power Plant which has high development potential, the following policies are required:
 - Large-Scale and Communal Solar Power Plant Development: Prioritizing solar power plant development in locations with high solar energy potential by simplifying licensing and providing incentives.
 - Building solar PV factories in Indonesia through cooperation and incentive schemes.
3. Regarding the use of energy which is still mainly centered in Java and Bali, in the next 10 years it is necessary to increase the development of regions outside Java and Bali, especially industrial development which can accelerate the increase in the regional economy.
4. To support the high development of electric vehicles, follow-up is needed, including:
 - Policies Related to Strengthening Local Production: Encourage investment in domestic battery and electric vehicle production to lower costs and increase competitiveness.
 - Distributed Charging Stations: The government and private sector need to accelerate the provision of Public Electric Vehicle Charging Stations (SPKLU) in strategic locations, such as toll roads, shopping centers, offices, and residential areas, as well as charging at home with affordable electricity rates.
 - Recycling and Waste Management Policy: A system to recycle electric vehicle batteries and manage their waste responsibly.

ATTACHMENT

ATTACHMENT 1

OUTLOOK SUMMARY

No	Energy Indicator	Unit	National Projection of REF Scenarios										
			2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
1	Final Energy Demand	MTOE	188	196	206	216	227	240	254	270	289	311	337
2	Final Energy Demand for Electricity	TWh	350	360	372	385	399	415	435	456	482	512	547
3	Primary Energy Supply	MTOE	269	279	291	304	318	333	351	371	394	423	456
4	Power Plant Capacity	GW	94	101	108	115	119	124	128	132	135	138	140
5	Electricity Production	TWh	371	383	396	410	425	442	463	486	514	546	584
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	810	835	873	906	945	989	1,041	1,102	1,168	1,257	1,359

No	Energy Indicator	Unit	National Projection of GREEN Scenarios										
			2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
1	Final Energy Demand	MTOE	188	194	200	207	215	224	233	245	259	275	295
2	Final Energy Demand for Electricity	TWh	349	374	400	429	460	493	530	566	608	657	713
3	Primary Energy Supply	MTOE	267	275	284	294	305	318	333	351	373	399	429
4	Power Plant Capacity	GW	93	104	124	136	143	151	165	175	190	199	213
5	Electricity Production	TWh	369	389	412	437	463	493	526	563	606	655	712
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	800	808	825	843	865	890	899	940	982	1,039	1,105

No	Energy Indicator	Unit	Sumatra Projection of REF Scenarios										
			2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
1	Final Energy Demand	MTOE	37	39	42	44	47	50	53	57	61	65	71
2	Final Energy Demand for Electricity	TWh	53	55	57	59	61	63	66	70	73	78	82
3	Primary Energy Supply	MTOE	52	54	56	59	62	66	70	74	79	85	91
4	Power Plant Capacity	GW	19	20	22	23	24	24	25	25	25	25	25
5	Electricity Production	TWh	59	61	63	65	67	70	73	77	81	85	90
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	124	127	138	145	151	158	167	177	188	201	216

No	Energy Indicator	Unit	Sumatra Projection of GREEN Scenarios										
			2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
1	Final Energy Demand	MTOE	38	39	40	41	43	44	46	48	51	54	58
2	Final Energy Demand for Electricity	TWh	53	57	62	68	74	81	88	97	108	120	134
3	Primary Energy Supply	MTOE	51	52	54	57	59	63	66	70	75	81	88
4	Power Plant Capacity	GW	19	20	25	26	26	27	30	32	37	39	43
5	Electricity Production	TWh	58	62	67	73	79	86	94	104	114	127	141
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	123	124	129	134	137	141	142	149	149	156	166

No	Energy Indicator	Unit	Java-Bali Projection of REF Scenarios										
			2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
1	Final Energy Demand	MTOE	109	113	118	123	129	135	142	149	158	169	181
2	Final Energy Demand for Electricity	TWh	231	237	245	254	263	274	287	302	320	341	365
3	Primary Energy Supply	MTOE	165	171	177	183	190	198	207	217	229	244	261
4	Power Plant Capacity	GW	50	54	57	62	65	68	71	74	77	80	82
5	Electricity Production	TWh	255	264	272	281	291	303	317	333	352	375	401
6	Greenhouse Gas Emissions	Mill. Ton CO _{2eq}	538	556	575	593	618	645	676	712	750	804	866

No	Energy Indicator	Unit	Java-Bali Projection of GREEN Scenarios										
			2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
1	Final Energy Demand	MTOE	109	111	114	117	121	124	129	134	141	148	157
2	Final Energy Demand for Electricity	TWh	229	240	252	265	279	294	310	329	350	374	402
3	Primary Energy Supply	MTOE	162	166	170	175	180	186	193	201	211	223	236
4	Power Plant Capacity	GW	50	53	55	57	59	61	65	70	76	81	86
5	Electricity Production	TWh	250	261	273	286	300	315	331	350	371	396	423
6	Greenhouse Gas Emissions	Mill. Ton CO _{2eq}	524	526	534	543	553	566	568	589	615	647	683

No	Energy Indicator	Unit	Kalimantan Projection of REF Scenarios										
			2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
1	Final Energy Demand	MTOE	23	25	26	27	29	31	33	36	39	43	48
2	Final Energy Demand for Electricity	TWh	14	14	15	15	16	17	18	18	19	21	22
3	Primary Energy Supply	MTOE	27	28	30	31	33	35	37	40	44	48	53
4	Power Plant Capacity	GW	5	6	7	7	7	8	8	8	8	8	8
5	Electricity Production	TWh	15	15	16	16	17	18	18	19	21	22	23
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	66	67	70	73	77	81	87	94	102	112	125

No	Energy Indicator	Unit	Kalimantan Projection of GREEN Scenarios										
			2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
1	Final Energy Demand	MTOE	21	22	22	23	24	25	26	28	30	32	35
2	Final Energy Demand for Electricity	TWh	15	16	18	20	22	25	28	32	37	42	49
3	Primary Energy Supply	MTOE	25	26	27	28	29	30	32	34	37	41	45
4	Power Plant Capacity	GW	5	7	11	16	19	21	24	26	27	29	31
5	Electricity Production	TWh	15	17	19	21	24	27	30	34	39	44	51
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	58	59	60	62	65	68	70	74	80	86	93

No	Energy Indicator	Unit	Sulawesi Projection of REF Scenarios										
			2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
1	Final Energy Demand	MTOE	12	13	14	15	17	18	19	21	23	25	28
2	Final Energy Demand for Electricity	TWh	38	39	41	42	43	45	47	49	52	55	58
3	Primary Energy Supply	MTOE	17	18	19	21	22	24	26	28	30	33	37
4	Power Plant Capacity	GW	11	11	12	12	13	13	13	14	14	14	14
5	Electricity Production	TWh	27	29	30	31	33	35	37	39	42	45	49
6	Greenhouse Gas Emissions	Mill. Ton CO _{2eq}	55	58	61	65	69	73	78	85	92	101	112

No	Energy Indicator	Unit	Sulawesi Projection of GREEN Scenarios										
			2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
1	Final Energy Demand	MTOE	14	15	17	18	20	22	23	25	27	29	32
2	Final Energy Demand for Electricity	TWh	38	45	52	59	66	74	81	85	89	94	100
3	Primary Energy Supply	MTOE	20	21	22	24	25	27	29	31	34	38	42
4	Power Plant Capacity	GW	11	13	16	18	19	20	22	22	23	24	26
5	Electricity Production	TWh	31	33	36	39	41	44	48	52	56	61	67
6	Greenhouse Gas Emissions	Mill. Ton CO _{2eq}	65	68	71	74	78	82	85	92	100	109	118

No	Energy Indicator	Unit	Nusra Projection of REF Scenarios										
			2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
1	Final Energy Demand	MTOE	2	2	3	3	3	3	3	3	4	4	4
2	Final Energy Demand for Electricity	TWh	6	6	6	6	6	6	7	8	8	8	8
3	Primary Energy Supply	MTOE	4	4	4	4	4	4	5	5	5	5	6
4	Power Plant Capacity	GW	2	2	2	2	2	2	2	2	2	2	2
5	Electricity Production	TWh	6	6	6	6	7	7	8	8	8	9	9
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	11	11	11	11	12	12	13	13	14	15	15

No	Energy Indicator	Unit	Nusra Projection of GREEN Scenarios										
			2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
1	Final Energy Demand	MTOE	3	3	3	3	3	3	3	4	4	4	4
2	Final Energy Demand for Electricity	TWh	6	6	7	7	8	8	10	10	11	12	12
3	Primary Energy Supply	MTOE	4	4	4	5	5	5	5	6	6	6	7
4	Power Plant Capacity	GW	2	3	7	8	9	9	10	11	12	12	12
5	Electricity Production	TWh	5	7	7	8	8	9	10	11	12	12	13
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	12	12	10	10	10	10	11	11	11	11	11

No	Energy Indicator	Unit	Maluku Projection of REF Scenarios										
			2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
1	Final Energy Demand	MTOE	1	1	2	2	2	2	2	2	3	3	3
2	Final Energy Demand for Electricity	TWh	6	6	7	7	7	7	7	7	7	8	8
3	Primary Energy Supply	MTOE	3	3	3	3	3	3	4	4	4	4	5
4	Power Plant Capacity	GW	5	6	6	6	6	6	6	6	6	6	6
5	Electricity Production	TWh	7	7	7	7	7	7	7	7	8	8	8
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	10	10	10	11	11	12	12	13	14	15	16

No	Energy Indicator	Unit	Maluku Projection of GREEN Scenarios										
			2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
1	Final Energy Demand	MTOE	2	2	2	2	3	3	3	4	4	5	5
2	Final Energy Demand for Electricity	TWh	6	7	7	7	7	8	8	8	9	9	10
3	Primary Energy Supply	MTOE	3	3	4	4	4	5	5	5	6	7	7
4	Power Plant Capacity	GW	5	6	7	7	8	8	8	8	9	9	9
5	Electricity Production	TWh	7	7	7	7	7	8	8	8	9	10	10
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	12	12	13	14	15	16	17	18	20	22	25

No	Energy Indicator	Unit	Papua Projection of REF Scenarios										
			2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
1	Final Energy Demand	MTOE	2	2	2	2	2	2	2	2	2	2	2
2	Final Energy Demand for Electricity	TWh	2	2	2	2	3	3	3	3	3	3	3
3	Primary Energy Supply	MTOE	2	2	2	2	2	3	3	3	3	3	3
4	Power Plant Capacity	GW	1	2	2	2	2	2	2	2	2	2	2
5	Electricity Production	TWh	2	3	3	3	3	3	3	3	3	3	3
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	7	7	7	7	7	7	7	8	8	8	9

No	Energy Indicator	Unit	Papua Projection of GREEN Scenarios										
			2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
1	Final Energy Demand	MTOE	2	2	2	2	2	2	2	2	2	2	3
2	Final Energy Demand for Electricity	TWh	2	3	3	3	3	4	4	4	5	5	5
3	Primary Energy Supply	MTOE	2	3	3	3	3	3	3	3	3	4	4
4	Power Plant Capacity	GW	1	2	4	4	5	5	5	6	6	6	6
5	Electricity Production	TWh	3	3	3	3	4	4	4	4	5	5	6
6	Greenhouse Gas Emissions	Mill. Ton CO ₂ eq	8	8	7	7	7	7	7	7	7	8	8

ATTACHMENT 2

CONVERSION FACTOR

Energy	Original Unit	Multiplier Factor to BOE (Barrel Oil Equivalent)
Coal		
Antrasit	Ton	4.9893
Imported Coal	Ton	4.2766
Kalimantan Coal	Ton	4.2000
Ombilin Coal	Ton	4.8452
Tanjung Enim Coal	Ton	3.7778
Lignite	Ton	3.0649
Riau Peat	Ton	2.5452
Briquette	Ton	3.5638
Biomass		
Charcoal	Ton	4.9713
Firewood	Ton	2.2979
Gas	MSCF	0.1796
Gas Products		
City Gas	Thousand Kcal	0.0007
CNG	Thousand Kcal	0.0007
LNG	Ton	8.0532
LNG	MMBTU	0.1796
LPG	Ton	8.5246

Energi	Satuan Asli	Faktor Pengali ke BOE (Barrel Oil Equivalent)
Oil		
Condensate	Barrel	0.9545
Crude Oil	Barrel	1.0000
Oil Fuel		
Aviation Gasoil (Avgas)	Kilo Liter	5.553
Aviation Turbine Gas (Avtur)	Kilo Liter	5.8907
Super TT gasoline	Kilo Liter	5.8275
Premix gasoline	Kilo Liter	5.8275
Premium gasoline	Kilo Liter	5.8275
Kerosene	Kilo Liter	5.9274
Diesel oil	Kilo Liter	6.4871
MDF	Kilo Liter	6.6078
FO	Kilo Liter	6.9612
Oil Products		
Other Oil Products	Barrel	1.0200
Refinery Fuel		
Refinery Fuel Gas (RFG)	Barrel	1.6728
Refinery Fuel Oil (RFO)	Barrel	1.1236
Feed Stock	Barrel	1.0423
Power Plant	MWh	0.6130

Source: HEESI, 2022 or Energy Balance 1990-1994, Department of Mining and Energy

ATTACHMENT 3

EMISSION FACTOR

No.	Energy Type	Net Energy Content	% Carbon Content	CO ₂ (Kg CO ₂ /TJ)
1	Gas	0	73.4	57,600
2	Gasoline	5.8	84.6	73,290
3	Avtur	5.9	85	71,348
4	Kerosene	5.9	85	72,430
5	Diesel	6.5	86.5	76,000
6	LPG	5.6	82	59,500
7	Coal	4	74.6	100,000
8	Briquette	3.5	74.6	100,000
9	Avgas	5.6	84.6	71,348
10	Fuel oil	7	86.5	77,900
11	DME	5.6	82	57,600



ATTACHMENT 4

GLOSSARY

Avgas

A type of aviation fuel used in piston engine aircraft.

Avtur (Aviation Turbine Fuel)

A type of aviation fuel designed for gas turbine engines-aircraft.

Coal

Sedimentary rocks that can burn, are formed from organic deposits, mainly plant remains and are formed through the process of coalescence.

Biomass

Renewable energy from organic materials such as animals and plants.

Biofuel

Any fuel, whether solid, liquid or gas, that is produced from organic materials which may be derived directly from plants or indirectly from industrial, commercial, domestic or agricultural waste.

B30

The result of mixing diesel oil with 30% vegetable oil derived from palm oil or Crude Palm Oil (CPO).

NRE (Renewable Energy)

Energy that comes from natural processes that is produced continuously and sustainably without having to wait for millions of years like fossil energy.

Carbon Emission

The remaining results of fuel combustion in internal combustion engines, external combustion engines, or jet engines that are released through the engine exhaust system.

Final Energy

Energy that can be directly consumed by end users.

Primary Energy

The energy that is first consumed in an area, both which comes directly or indirectly from nature.

Gas

The results of natural processes in hydrocarbons which, under conditions of atmospheric pressure and temperature, are in the form of a gas phase obtained from the oil and gas mining process.

Industry

A field or economic activity related to the processing/manufacture of raw materials or finished goods in factories using skills and labor and the use of tools in the field of processing agricultural products and their distribution as the main activity.

KEN (*Kebijakan Energi Nasional*/National Energy Policy)

Energy management policies based on just, sustainable and environmentally friendly principles to create national energy independence and security.

Commercial

An activity carried out with the aim of gaining profit, generally in the form of activities related to commerce.

LEAP (Low Emission Analysis Platform)

An energy planning simulation model that is capable of analyzing energy from demand to supply in an integrated manner.

Oil

The results of natural processes in hydrocarbons which, under conditions of atmospheric pressure and temperature, are in the liquid or solid phase.

Power Plant

A collection of equipment and machines used to generate electricity through the process of transforming energy from various energy sources.

Region

Regional division based on geographic proximity used in energy modeling (LEAP).

Household

Consists of one or several people who live in the same residence which include one family or group of people.

Other Sectors

Several activities that correlate with supporting commercial businesses consist of agricultural, construction and mining activities.

Scenario

Conditions or circumstances that may occur based on the assumptions input into energy modeling.

Transportation

One part of the goods and passenger distribution network that has developed very dynamically and plays a role in supporting political, economic, socio-cultural as well as defense and security development.

NZE (Net Zero Emission)

Conditions where the amount of carbon emissions released into the atmosphere does not exceed the amount of emissions that the earth can absorb.

RUEN (*Rencana Umum Energi Nasional*/National Energy General Plan)

The policy by the central government regarding the national energy management plan as the elaboration and implementation of the National Energy Policy which is cross-sectoral in nature to achieve the National Energy Policy targets.

Gross Domestic Product

The value of all goods and services produced by a country in a certain period.

Projection

Projected conditions that may occur at a certain time based on the assumptions used.

ATTACHMENT 5

ABBREVIATIONS

APBN	: <i>Anggaran Pendapatan dan Belanja Negara</i> (State Budget)
APK DEN	: <i>Anggota Pemangku Kepentingan Dewan Energi Nasional</i> (Members of National Energy Council from Stakeholders)
Bappenas	: <i>Badan Perencanaan Pembangunan Nasional</i> (National Development Plan Agency)
BBM	: <i>Bahan Bakar Minyak</i> (Fuel)
BBN	: <i>Bahan Bakar Nabati</i> (Biofuel)
BOE	: Barrel Oil Equivalent
BPP	: <i>Biaya Pokok Penyediaan</i> (Basic Cost of Supply)
BPS	: <i>Badan Pusat Statistik</i> (Statistics Indonesia)
BPH	: <i>Badan Pengatur Hilir</i> (Downstream Task Force)
BPPT	: <i>Badan Pengkajian dan Penerapan Teknologi</i> (Technology Assessment and Application Agency)
BTS	: Base Transceiver Station
CCS	: Carbon Capture and Storage
CCUS	: Carbon Capture, Utilization and Storage
CO ₂	: Carbon Dioxide
CO ₂ eq	: Carbon Dioxide equivalent
Covid-19	: Corona Virus Disease 2019
DEN	: <i>Dewan Energi Nasional</i> (National Energy Council)
Ditjen Migas	: <i>Direktorat Jenderal Minyak dan Gas Bumi</i> (Directorate General of Oil and Gas)
DME	: Dimethyl Ether
EV	: Electric Vehicle

EBT	: <i>Energi Baru Terbarukan</i> (New Renewable Energy)
EBTKE	: <i>Energi Baru Terbarukan dan Konservasi Energi</i> (New Renewable Energy and Energy Conservation)
GIZ	: Deutsche Gesellschaft für Internationale Zusammenarbeit
GRK	: <i>Gas Rumah Kaca</i> (Green House Gas)
GSEN	: <i>Grand Strategi Energi Nasional</i> (National Energy Grand Strategy)
Gt	: Giga tonne
GW	: Giga Watt
Gwe	: Giga Watt electric
GWh	: Giga Watt hour
HBA	: <i>Harga Batubara Acuan</i> (Coal Reference Price)
HEESI	: Handbook Of Energy & Economic Statistics Of Indonesia
H ₂	: <i>Hidrogen</i> (Hydrogen)
IAEA	: International Atomic Energy Agency
IEA	: International Energy Agency
IKN	: <i>Ibu Kota Nusantara</i> (Capital City)
INAFHE	: Indonesian Association for Fuel Cell and Hydrogen Energy
ITB	: <i>Insitut Teknologi Bandung</i>
Jargas	: <i>Jaringan Gas</i> (City gas)
JBKP	: <i>Jenis BBM Khusus Penugasan</i> (Certain Fuel for Assignment)
JBT	: <i>Jenis BBM Tertentu</i> (Certain Fuel)
KAI	: <i>Kereta Api Indonesia</i> (Indonesia railway)
KBLBB	: <i>Kendaraan Bermotor Listrik Berbasis Baterai</i> (Battery based electric vehicles)
KEN	: <i>Kebijakan Energi Nasional</i> (National Energy Policy)
KESDM	: <i>Kementerian Energi dan Sumber Daya Mineral</i> (Ministry of Energy and Mineral Resources)
KOICA	: Korea International Cooperation Agency
kW	: kilo Watt
kWh	: kilo Watt hour
LEAP	: Low Emissions Analysis Platform
LPEM	: <i>Lembaga Penyelidikan Ekonomi dan Masyarakat</i> (Institute for Research on Economics and Society)
LPG	: Liquefied Petroleum Gas
LNG	: Liquefied Natural Gas
LSM	: <i>Lembaga Swadaya Masyarakat</i> (NGO)
LTSHE	: <i>Lampu Tenaga Surya Hemat Energi</i> (Energy Saving Solar Lights)
MBSD	: Million Barel Steam Per Day
MMBTU	: Million British Thermal Unit

MT	: Metrik Ton
MTPA	: Metric Ton per Annum
MMPA	: Million Metric Ton per Annum
MVA	: Mega Volt Ampere
MW	: Mega Watt
MWh	: Mega Watt hour
NEPIO	: Nuclear Energy Program Implementing Organization
NDC	: Nationally Determined Contributions
NTB	: <i>Nusa Tenggara Barat</i> (West Nusa Tenggara)
NTT	: <i>Nusa Tenggara Timur</i> (East Nusa Tenggara)
Nusra	: <i>Nusa Tenggara</i>
NZE	: Net Zero Emission
OEI	: <i>Outlook Energi Indonesia</i> (Indonesia Energy Outlook)
O ₂	: Oksigen (Oxygen)
PSO	: Public Service Obligation
PLN	: <i>Perusahaan Listrik Negara</i> (Electricity State Own Enterprise)
Perpres	: <i>Peraturan Presiden</i> (Presidential Regulation)
PLTA	: <i>Pembangkit Listrik Tenaga Air</i> (Hydro Power Plant)
PLTB	: <i>Pembangkit Listrik Tenaga Bayu</i> (Wind Power Plant)
PLTD	: <i>Pembangkit Listrik Tenaga Diesel</i> (Diesel Power Plant)
PLTG	: <i>Pembangkit Listrik Tenaga Gas</i> (Gas Power Plant)
PLTM	: <i>Pembangkit Listrik Tenaga Minihidro</i> (Minihydro Power Plant)
PLTMH	: <i>Pembangkit Listrik Tenaga Mikrohidro</i> (Microhydro Power Plant)
PLTN	: <i>Pembangkit Listrik Tenaga Nuklir</i> (Nuclear Power Plant)
PLTP	: <i>Pembangkit Listrik Tenaga Panas Bumi</i> (Geothermal Power Plant)
PLTS	: <i>Pembangkit Listrik Tenaga Surya</i> (Solar Power Plant)
PLTU	: <i>Pembangkit Listrik Tenaga Uap</i> (Coal Fired Power Plant)
PP	: <i>Peraturan Pemerintah</i> (Government Regulation)
Pusdatin	: <i>Pusat Data dan Teknologi Informasi</i> (Center for Data and Technology Information)
Puslitbang	: <i>Pusat Penelitian dan Pengembangan</i> (Research and Development Center)
PUPR	: <i>Pekerjaan Umum dan Perumahan Rakyat</i> (Public Works and Housing)
RUPTL	: <i>Rencana Usaha Penyediaan Tenaga Listrik</i> (Power Supply Business Plan)
RUEN	: <i>Rencana Umum Energi Nasional</i> (National Energy General Plan)

RON	: Research Octane Number
Setjen/SJ DEN	: <i>Sekretariat Jenderal Dewan Energi Nasional</i> (Secretariat General of the National Energy Council)
SPBE	: <i>Stasiun Pengisian Bulk Elpiji</i> (LPG Filling Station)
SPBU	: <i>Stasiun Pengisian Bahan Bakar Umum</i> (Fuel Filling Stations)
SPBKLU	: <i>Stasiun Penukaran Baterai Kendaraan Listrik Umum</i> (Public Electric Vehicle Battery Swap Station)
SPKLU	: <i>Stasiun Pengisian Kendaraan Listrik Umum</i> (Public Electric Vehicle Charging Station)
SR	: <i>Sambungan Rumah</i> (Household Connection)
SWH	: Solar Water Heater
TOE	: Tonne of Oil Equivalent
TSCF	: Triliun Standard Cubic Feet
TWh	: Terra Watt hour
UI	: <i>Universitas Indonesia</i>
USC	: Ultra Super Critical
USD	: United States Dollar
VA	: Volt Ampere
3T	: <i>Tertinggal, Terdepan dan Terluar</i> (Disadvantages, Frontier and Outermost)



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